

INTRODUCTION

Carbohydrates carry out several roles in living organisms. Carbohydrates may be simplest as monosaccharides and complex as disaccharides, oligosaccharides and polysaccharides. Basically, complex carbohydrates are formed from condensation among varying number and types of monosaccharides as monomer sugar units. Polysaccharides serve as an energy reservoir such as starch and glycogen as well as structural constituents for example cellulose and chitin. Thus, carbohydrates and their derived compounds such as glycoconjugates are vital bio-molecules which perform chief functions in the providing immunity against many diseases, blood clotting, fertilization and development.

Carbohydrates, also known as "**saccharides**," are classified according to the number of single carbohydrate molecules in each chemical structure. Carbohydrate compounds having just one carbohydrate molecule are called "**monosaccharides**;" compounds with two carbohydrate molecules are called "**disaccharides**;" and those compounds containing more than two carbohydrate molecules are named "**polysaccharides**". All carbohydrates either are monosaccharides or can be hydrolyzed (broken down) into two or more monosaccharides.

For further understanding of these different classifications of carbohydrates, the monosaccharides and disaccharides can be grouped together and compared with the polysaccharides. This can be done because monosaccharides and disaccharides have certain common characters: For one, they are both water soluble. In addition, they have a sweet taste and a crystalline structure. The monosaccharides and disaccharides are called "**sugars**" and all share the suffix, "**-ose**," meaning *sugar*.

Polysaccharides, in contrast to mono- and disaccharides, are insoluble in water, do not taste sweet and do not form crystals. Also, they do not share a suffix and have no group name (such as sugars, in the case of mono- and disaccharides). They are sometimes called starches, but this is technically incorrect because there are many other classifications of polysaccharides besides starches (cellulose and glycogen being two and dextrin being another).

Carbohydrates Definition:-

Carbohydrates are the most abundant organic molecules in nature. They are organic substances composed essentially of C, H and O, with the general formula " $C_n(H_2O)_n$ ". They are defined as simple sugars or their polymers; simple sugars are polyhydroxy-aldehydes or polyhydroxy-ketones.

Biological Importance's of carbohydrates

❖ They are used as material for energy storage and production.

- **Starch and glycogen**, respectively in plants and animals, are stored carbohydrates from which glucose can be mobilized for energy production. Glucose can supply energy both fueling ATP synthesis (ATP, the cell's energy currency, has inside a phosphorylated sugar) and in the form of reducing power as NADPH.
- It should be noted that glucose, used as energy source, "burns" without yielding metabolic wastes, being turned in CO_2 and water, and of course releasing energy.
- Monosaccharides supply 3.74 kcal/g, disaccharides 3.95 kcal/g, while starch 4.18 kcal/g; on average it is approached to 4 kcal/g.

❖ They exert a protein-saving action:

if present in adequate amount in daily nourishment, the body does not utilize proteins for energy purpose, an anti-economic and "polluting" fuel because it will need to eliminate nitrogen (ammonia) and sulfur present in some amino acids.

❖ Their presence is necessary for the normal lipid metabolism.

More than 100 years ago Pasteur said: "Fats burn in the fire of carbohydrates". This idea continues to receive confirmations from the recent scientific studies. Moreover, excess carbohydrates may be converted in fatty acids and triglycerides (processes that occur mostly in the liver).

❖ Glucose is indispensable for the maintenance of the integrity of nervous tissue

(some central nervous system areas are able to use only glucose for energy production) and red blood cells.

❖ Two sugars, ribose and deoxyribose, are part of the bearing structure, respectively of the RNA and DNA and obviously find themselves in the nucleotide structure as well.

❖ They take part in detoxifying processes.

For example, at hepatic level glucuronic acid, synthesized from glucose, combines with endogenous substances, as hormones, bilirubin etc., and exogenous substances, as chemical or bacterial toxins or drugs, making them atoxic, increasing their solubility and allowing their elimination.

❖ **They are also found linked to many proteins and lipids.**

Within cells they act as **signals** that determine the metabolic fate or the intracellular localization of the molecules which are bound. On the cellular surface their presence is necessary for identification processes between cells that are involved e.g. in the recognition between spermatozoon and oocyte during fertilization, in the return of lymphocytes in the lymph nodes of provenance or still in the leukocyte adhesion to the lips of the lesion of a blood vessel.

❖ **Two homopolysaccharides, cellulose (the most abundant polysaccharide in nature) and chitin (probably, next to cellulose, the second most abundant polysaccharide in nature), serve as structural elements, respectively, in plant cell walls and exoskeletons of nearly a million species of arthropods (e.g. insects, lobsters, and crabs).**

❖ **Heteropolysaccharides provide extracellular support for organisms of all kingdoms:**

In bacteria, the rigid layer of the cell wall is composed in part of a heteropolysaccharide contained two alternating monosaccharide units while in animals the extracellular space is occupied by several types of heteropolysaccharides, which form a matrix with numerous functions, as hold individual cells together and provide protection, support, and shape to cells, tissues, and organs.

Classification of Carbohydrates

Three Building Stocks of Carbohydrates: Sugars, Starches and Fiber

Carbohydrates are usually classified into three broad categories: sugars, starches, and fiber.

- ❖ **Sugars**: are simple carbohydrates like glucose, fructose and lactose. They cause a rapid rise in blood glucose levels.
- ❖ **Starches**: are complex carbohydrates that are composed of several molecules of glucose. They cause a less rapid rise in blood glucose levels. On the other hand, resistant starch and fiber are not digested or broken down in the small intestine, but these have many positive health effects.

- ❖ **Fiber**: is a non-digestible complex carbohydrate. Our gut does not possess the enzymes needed to break apart the links between sugar units. Undigested fiber travels through our gut and while doing so, provides health benefits.

Physiological Classification of Carbohydrates

The physiologic classification includes:

- ❖ **Simple Carbohydrates**

These include sugars like monosaccharides, disaccharides and oligosaccharides like trisaccharides and tetrasaccharides. Simple carbohydrates are easily digested basic sugars that offer little health value for the body when taken in large amounts. Foods that are high in simple sugars and low in fiber increase your blood sugar levels. Although fruits and vegetables contain simple carbohydrates, they are healthier than processed foods like cookies and cakes because they contain fiber. This changes the way that your body processes the sugars because fiber slows down their absorption.

It is therefore important to limit your intake of simple sugars from processed foods like: [Candy, Soda, Artificially sweetened syrups, Table sugar, White rice, bread, or pasta, Pastries and other desserts].

- ❖ **Complex Carbohydrates**

The polysaccharides are complex carbohydrates that are considered good for health because it takes more time for the body to break them down. They usually have a low glycemic load, meaning that you get lower amounts of sugar, which is released at a slower rate, producing small increases in blood sugar levels, instead of the peaks and valleys characteristic of simple sugars.

Choose complex carbohydrates over foods with simple carbohydrates by making simple substitutions in your meals. For example, choose brown rice instead of your usual white rice, or eat whole-wheat bread or pasta instead of white bread or pasta.

Chemical Classification of Carbohydrates

The chemical classification includes monosaccharides, oligosaccharides and polysacchares.

Classification and Nomenclature						
Carbohydrates						
Monosaccharide		Oligosaccharide			Polysaccharide	
Functional group	Number of carbon atoms	Di-saccharide	Tri-saccharide	Tetra-saccharide	Homopoly-saccharide	Heteropoly-saccharide
Aldoses e.g. Glucose	Trioses	Maltose	Raffinose	Stachyose	Starch	Hyaluronic acid
	Tetroses	Lactose			Dextrin	Heparin
Ketoses e.g. Fructose	Pentoses	Sucrose			Glycogen	Chondroitin sulfate
	Hexoses				Cellulose	Dermatan Sulfate
	Heptoses				Inulin	Keratan Sulfate

❖ Monosaccharides

Monosaccharides are simple sugars, which possess a free ketone or aldehyde group. Being the simplest of sugars, they cannot be further hydrolyzed. Their chemical formula is " $C_nH_{2n}O_n$ " or " $C_n(H_2O)_n$ " Monosaccharides are classified into trioses, tetroses, pentoses, etc., and as ketoses or aldoses, depending on their ketone or aldehyde group.

Examples include glucose, fructose, galactose, glycerose, ribose, and ribulose.

❖ Oligosaccharides

“Oligo” meaning few, oligosaccharides are sugars that break down into two to 10 molecules of monosaccharides when hydrolyzed.

An oligosaccharide that yields two monosaccharide molecules on hydrolysis is a disaccharide, while those that break down into three or four monosaccharides are called trisaccharides, tetrasaccharides, and so on. Disaccharides have a chemical formula of " $C_n(H_2O)_{n-1}$ " while trisaccharides and others are " $C_n(H_2O)_{n-2}$," etc.

Oligosaccharide examples include sucrose, maltose, lactose, raffinose, and stachyose.

❖ Polysaccharides:-

“Poly” meaning many, polysaccharides are compound molecules that yield more than ten monosaccharide molecules on hydrolysis. They are also classified depending on the type of molecules hydrolyzed. These include homopolysaccharides (with several monosaccharides of one type) or heteropolysaccharides (with different types of monosaccharides). " $(C_6H_{10}O_5)_x$ " is their chemical formula.

Polysaccharide examples include starch, cellulose, pectin, glycogen, inulin, and hyaluronic acid.

Monosaccharides

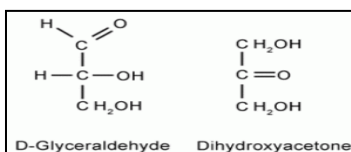
These are the only sugars that can be absorbed and utilized by the body. Most of the oxygen atoms in monosaccharides are found in hydroxyl groups, but one of them is part of a carbonyl group. The position of the carbonyl group can be used to categorize the sugars:

- If the sugar has an "**aldehyde group**", meaning that the carbonyl C is the last one in the chain, it is known as an "**aldose**".
- If the carbonyl C is internal to the chain, so that there are other carbons on both sides of it, it forms a "**ketone**" group and the sugar is called a "**ketose**".

Monosaccharides are categorized by the number of carbon & whether an aldehyde / ketone.

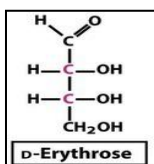
❖ Trioses:-

Simplest monosaccharide contains three carbon atoms (eg.. Glyceraldehyde & Dihydroxyacetone)



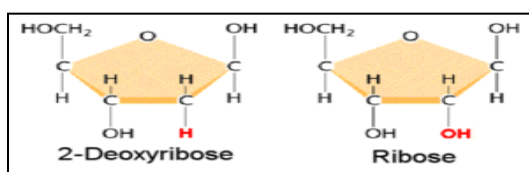
❖ Tetroses:-

Contain four carbons. They have either a functional aldehyde group in position 1 or functional ketone group in position 2. (eg: D-Erythrose).



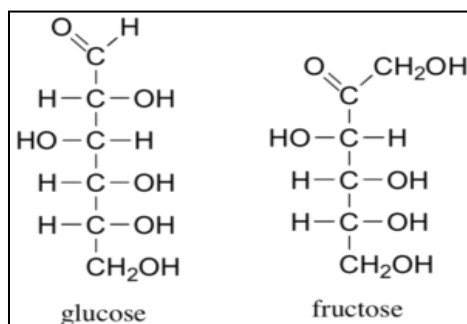
❖ Pentoses:-

Contain five carbons. Aldopentoses have an functional aldehyde group at position 1 & ketopentoses have an functional ketone group at position 2/3 (eg:. Xylose, Ribulose, Arabinose, Deoxyribose).

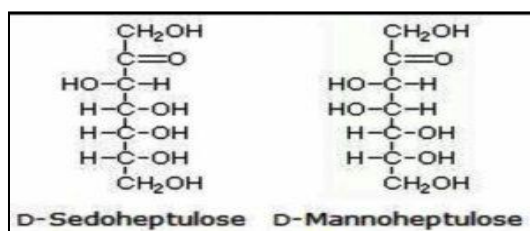


❖ Hexoses:-

Contain six carbons. Aldohexoses having functional aldehyde group at position 1 & ketohexoses having functional ketone group at position 2. (e.g. Glucose, Fructose).

❖ Heptoses:-

Contain seven carbons. Either aldohexoses have a functional aldehyde group in position 1 or ketohexoses have a functional ketone group in position 2. (e.g. sedoheptulose, Mannoheptulose).

✚ Stereoisomerism:-

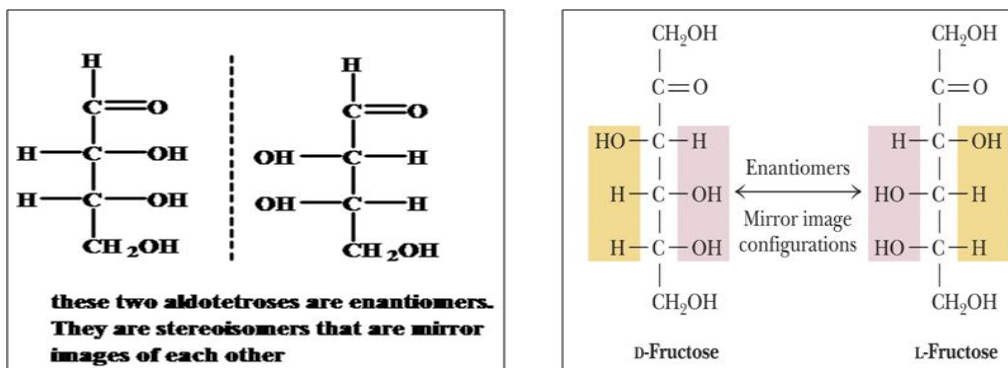
Isomers are compounds with identical molecular formulas. Isomers can be categorized into the two different groups of constitutional isomers or stereoisomers.

Constitutional isomers have the same molecular formula but a different molecular framework (different bonding constitution). Because constitutional isomers have different bonding constitutions, they are different molecules. This means that constitutional isomers have different physical and chemical properties. Ethanol $\text{CH}_3\text{CH}_2\text{OH}$ and dimethyl ether CH_3OCH_3 are constitutional isomers. Both have the same molecular formula ($\text{C}_2\text{H}_6\text{O}$) but differ in how the atoms are connected.

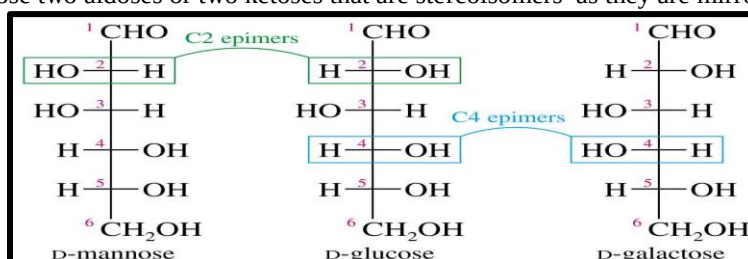
Stereoisomers are molecules containing the same atoms bonded identically but the bonded atoms are oriented differently in space. That is to say, they have identical bonding constitutions but differ in how the atoms are oriented in the space around the atoms to which they are bonded.

Stereoisomers can be further separated into the two categories of diastereomers and enantiomers. One type of diastereomers (or geometric stereoisomers) differ by "cis" and "trans" orientations.

Enantiomers are a class of stereoisomers related like an object and its mirror image. Enantiomers differ in their "handedness" as the left hand and right hand are related. Enantiomers are a pair of mirror image molecules that cannot be superimposed on each other. Superimposed suggests that two mirror image molecules can be mentally merged into one object as they are brought together.

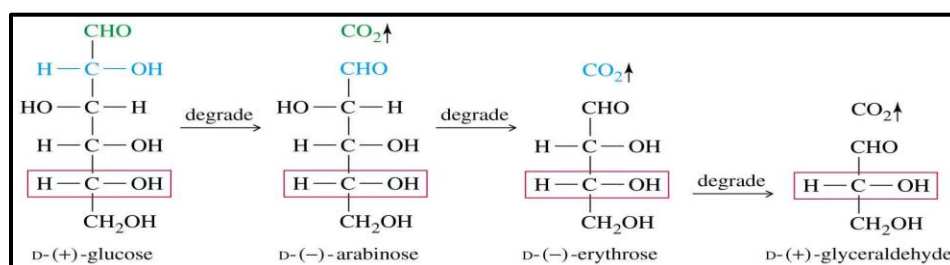


Enantiomers : are Those two aldoses or two ketoses that are stereoisomers as they are mirror images for each other



Epimers : Sugars that differ only in their stereochemistry at a single carbon.

There are two prominent "handed" biologically important molecules. The D- sugars and L- amino acids. The designations of D- and L- refer to how the pair of enantiomers differ in their bonding configurations. In biochemistry, D is a symbol used as a prefix to indicate the spatial configuration of certain organic compounds with asymmetric carbon atoms. It is used if an organic compound has a configuration about an asymmetric carbon atom (chiral center) analogous to that of D-glyceraldehyde (the arbitrarily chosen standard), in which the hydroxy (OH) functional group is on the right side of the asymmetric carbon atom.



- D - sugars can be degraded to the dextrorotatory (+) form of glyceraldehyde.
- L - sugars can be degraded to the levorotatory (-) form of glyceraldehyde.

The term "chirality" refers to the "handedness" of a molecule. Chiral molecules have a chiral center and these pair of molecules cannot be superimposed. A chiral center is an atom with four different substituents. A carbon atom that has four different groups bonded to it is called asymmetric carbon or a chiral carbon. On the other hand, (humor!) achiral molecules (molecules "without handedness") can be superimposed.

Enantiomers are identical in most physical and chemical properties such as: melting point, boiling point, density, and chemical reactions typical for the functional groups present in the molecule.

However, there are two physical properties which permit discernment of chirality:

- Chiral molecules differ in their interaction with plane polarized light. (Chiral molecules are sometimes called optical isomers.) A polarimeter is an instrument that allows plane polarized light to pass through aqueous solution of the molecule. The (+) isomer rotates plane polarized light clockwise. The (-) isomer rotates plane polarized light counterclockwise. Achiral molecules do not rotate polarized light in either direction. Racemic mixtures contain equal mix of (+) and (-) isomers. Racemic mixtures show NO rotation of polarized light.
- Chiral also molecules differ in their interaction with other chiral compounds. Chiral molecules specifically recognize other chiral molecules. For example, L-amino acid protein enzyme (chiral molecule)

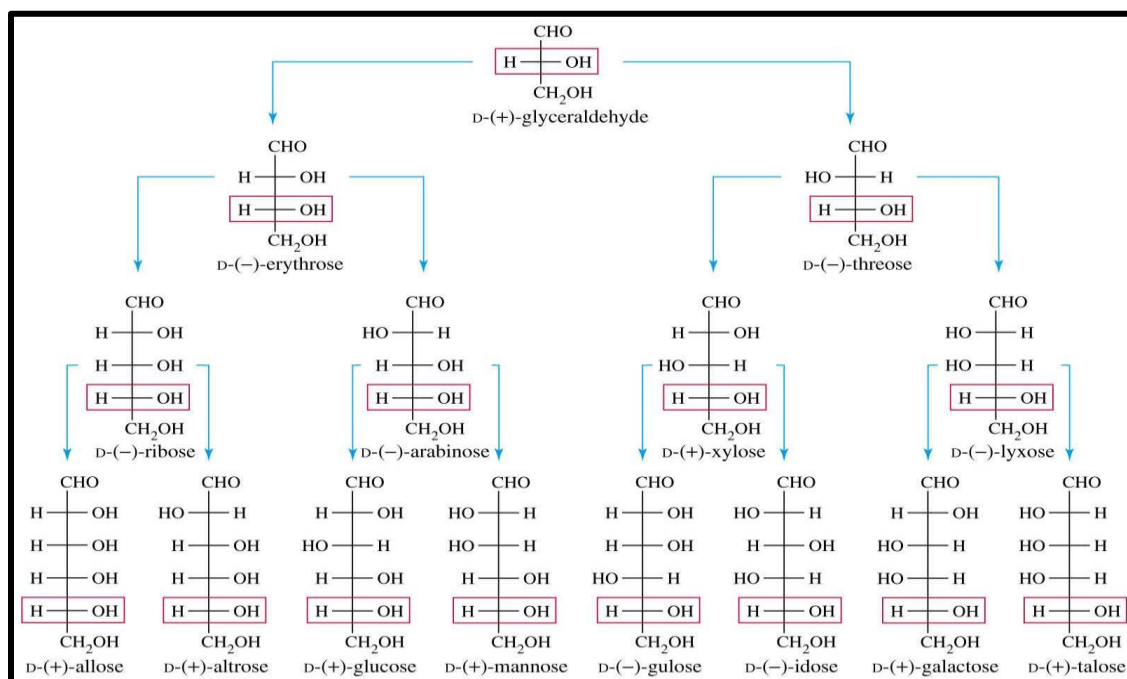
How many stereoisomers can a molecule have?

The number of possible stereoisomers depends upon the number of chiral centers in the molecule. Van't Hoffs rule states: number of stereoisomers = 2^n , where n = number of chiral centers. For example, a molecule with 2 chiral centers can have 4 stereoisomers.

Structural Representation of Sugars:-

- ❖ **Fisher projection:** straight chain representation.(i.e: a standard method for depicting the three-dimensional arrangement of atoms on a page).
 - Tetrahedral carbon atoms are represented by two crossed lines.
 - The horizontal lines represent bonds coming out of the page.

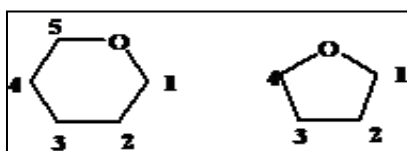
- The vertical lines represent bonds going into the page.
- Carbonyl carbon is placed at or near the top in Fischer projections.
- Fischer projections can be rotated 180° without changing their meaning, but not by 90° or 270° .
- Carbohydrates with more than one stereogenic center are shown by stacking the centers on top of one another, with the carbonyl carbon again placed at or near the top. D-sugars have the stereogenic carbon farthest from the carbonyl with the hydroxyl group written on the right of the molecule.



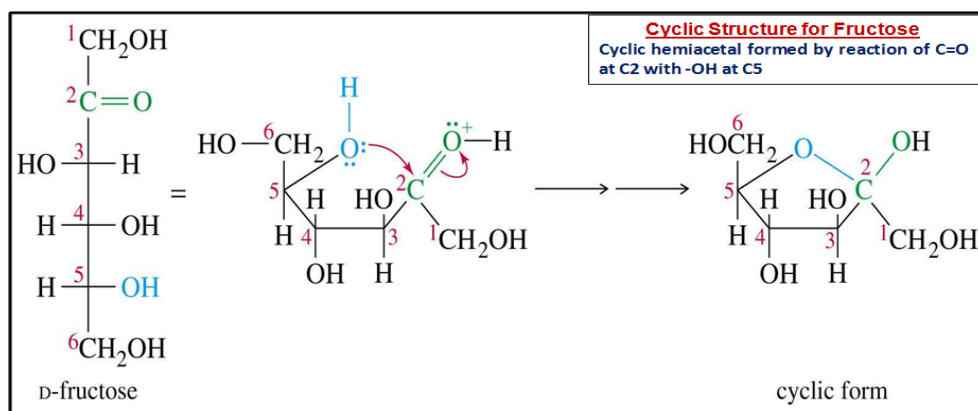
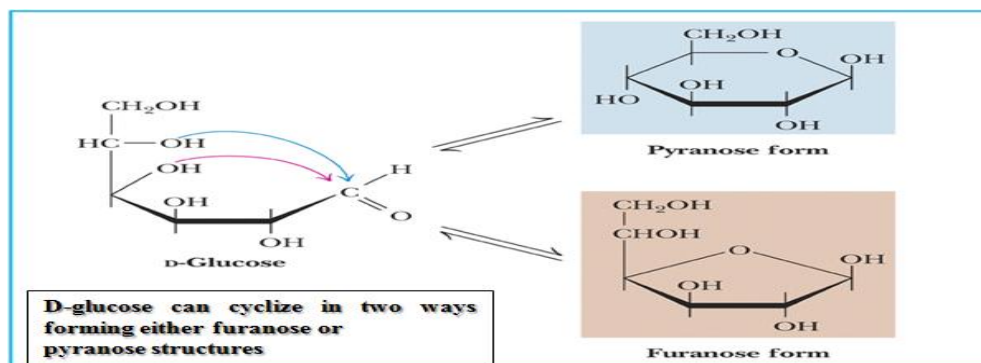
❖ **Haworth projection:** simple ring in perspective. A method to depict ring structures (flat) arrange the ring with O in the back (or back-right)

Rules for drawing Haworth projections:

- Draw either a six or 5-membered ring including oxygen as one atom.
- Most aldohexoses are six-membered.
- Aldotetroses, aldopentoses, ketohexoses are 5 membered.



- Next number the ring clockwise starting next to the oxygen
- If the substituent is to the right in the Fisher projection, it will be drawn down in the Haworth projection (Down-Right Rule)



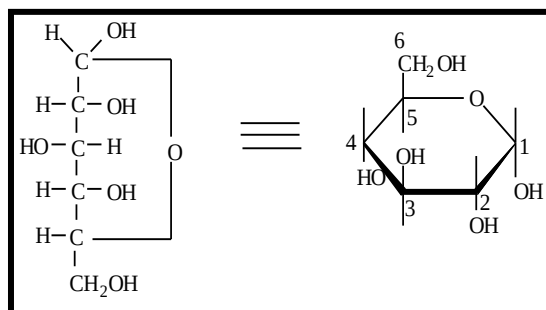
Anomeric Carbon : The carbon atom which is involved in hemiacetal or acetal formation.

- For D-sugars the highest numbered carbon (furthest from the carbonyl) is drawn up. For L-sugars, it is drawn down.
- For D-sugars, the OH group at the anomeric position is drawn down for a and up for b. For L-sugars a is up and b is down
- ❖ **Conformational representation:** chair and boat configurations

Monosaccharides

Three monosaccharides are particularly important in the study of nutritional science: glucose, fructose and galactose.

❖ Glucose (also known as dextrose or grape sugar)

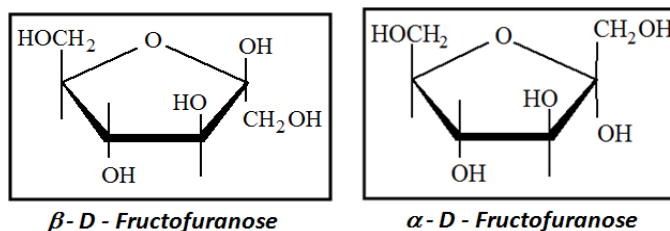


This monosaccharide is the most important carbohydrate in human nutrition because it is the one that the body fuses directly to supply its energy needs. Glucose is formed from the hydrolysis of di- and polysaccharides, including starch, dextrin, maltose, sucrose and lactose; from the monosaccharide fructose largely during absorption; and from both fructose and galactose in the liver during metabolism.

Glucose is the carbohydrate found in the bloodstream, and it provides an immediate source of energy for the body's cells and tissues. Glucose is also formed when stored body carbohydrate (glycogen) is broken down for use.

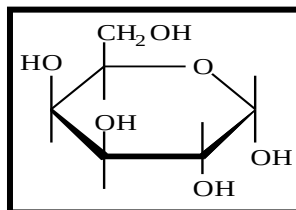
In the plant world, glucose is widely distributed. It is found in all plants and in the sap of trees. Fruits and vegetables are wholesome food sources of glucose. It is also present in such unwholesome (to humans) substances as molasses, honey and corn syrup.

❖ Fructose (also known as levulose or fruit sugar)



Fructose, a monosaccharide, is very similar to another monosaccharide, galactose. These two simple sugars share the same chemical formula; however, the arrangements of their chemical groups along the chemical chain differ. Fructose is the sweetest of all the sugars and is found in fruits, vegetables and the nectar of flowers, as well as in the unwholesome (to humans) sweeteners, molasses and honey. In humans, fructose is produced during the hydrolysis of the disaccharide, sucrose.

❖ Galactose



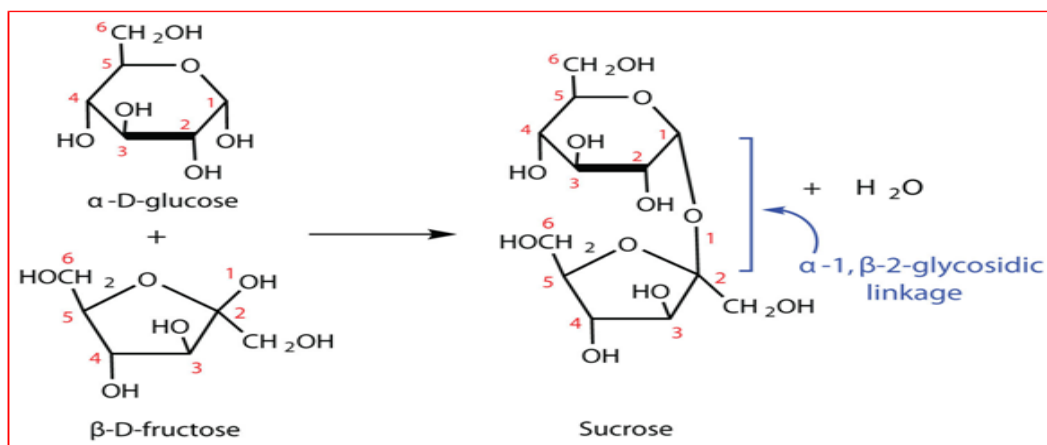
Galactose differs from the other simple sugars, glucose and fructose, in that it does not occur free in nature. It is produced in the body in the digestion of lactose (milk sugar), a disaccharide.

Disaccharides

Disaccharides (di- = “two”) formed from dehydration reaction between two monosaccharide units. During dehydration reaction, a hydroxyl group of one monosaccharide unit reacts with one hydrogen atom attached to the anomeric carbon of second monosaccharide unit, liberating water (H_2O) molecule and forms an O-glycosidic linkage.

In general rule, the numbering of carbon atoms in a monosaccharide are starts from the carbon just near to the oxygen atom of carbonyl group. For example, an O-glycosidic bond is formed between C1 in glucose and C2 in fructose that results in formation of sucrose, a disaccharides. Glycosidic linkages are easily broken through an acid but resist to cleavage through base. Therefore, disaccharides may be hydrolyzed to produce their free monosaccharide units through boiling with an acid.

Like monosaccharides, all common disaccharides are named by adding the suffix “-ose”. Sucrose, lactose, and maltose are the most abundant disaccharides. Sucrase, lactase and maltase found on the outer surface of epithelial cells lining small intestine and catalyze hydrolysis of sucrose, lactose and maltose into their corresponding monosaccharide units.



Formation of sucrose from glucose and fructose as monomer unit.

❖ Sucrose:-

- **Systematic name:** D-glucopyranosyl-(1 $\leftrightarrow$ 2)- β -D-fructofuranoside
- **Molecular formula:** $C_{12}H_{22}O_{11}$
- **Common name:** Table sugar, Saccharose

➤ **Structure:**

- It is composed of the monosaccharide unit glucose and fructose. The anomeric carbons of glucose and fructose as monomer unit are attached through glycosidic bond to form sucrose. The orientation of the glycosidic bond is α (alpha) at glucose and β (beta) at fructose.
- Sucrase are the enzymes located in the brush border of small intestine. Sucrase catalyzes the hydrolysis of sucrose to its constituent monosaccharide units fructose and glucose.
- Unlike maltose and lactose, sucrose has no free anomeric carbon since the anomeric carbons of both glucose and fructose units are participating in glycosidic linkage. Thus sucrose is a non-reducing disaccharide.

➤ **Importance:**

- The most abundant disaccharide is sucrose which is obtained commercially from sugarcane or sugar beet. Sucrose is best recognized as it is essential for nutrition of human and is only synthesized in plants.
- Sucrose is a mainly produced during photosynthesis in several plants. Sucrose is a principal form of sugar that is transported from leaves to other parts such as fruits and flower of the plant.
- Sucrose is a certainly assimilated macronutrient which serves a rapid source of energy to our body, stimulating a rapid increase in blood glucose after ingestion. Basically, pure sucrose is not part of a human balanced diet but it is included cautiously in the diet to make some foods tastier.
- Sucrose (in pure form) contains energy of 3.94 kilocalories (or 17 kilojoules) per gram of weight. If an excess of foods having high content of sucrose is consumed, displacement of essential nutrients from the diet could occur which may in turn cause an increased risk for chronic disease like defected glucose metabolism or *diabetes mellitus*.

❖ **Lactose :-**

- **Systemic name:** 4-O- β -D-galactopyranosyl-D-glucose
- **Abbreviated name:** Gal(β 1 $\rightarrow$ 4)Glc
- **Molecular formula:** C₁₂H₂₂O₁₁

- **Common name:** Milk sugar
- **Structure:**
 - Lactose is a disaccharide which made up of monosaccharide units, β -D-galactose and β -D-glucose linked through a $\beta(1\rightarrow4)$ glycosidic bond. Thus, its abbreviated name is Gal($\beta 1\rightarrow 4$)Glc.
 - Lactase in humans and β -galactosidase in bacteria involved in the hydrolysis of the lactose into its monosaccharide units, galactose and glucose.
 - Since the anomeric carbon atom of the glucose unit is free for oxidation reaction, lactose is said to be reducing disaccharide.
- **Importance:**
 - Lactose sugar naturally present in milk that means it is the only major carbohydrate having animal origin. Lactose forms 4.5 percent of milk of cow and 6.5 percent from human milk by weight.
 - A most of people do not contain lactase enzyme that digests lactose makes difficult for them to digest milk since upon ingestion causes gaseousness, cramping and diarrhea.

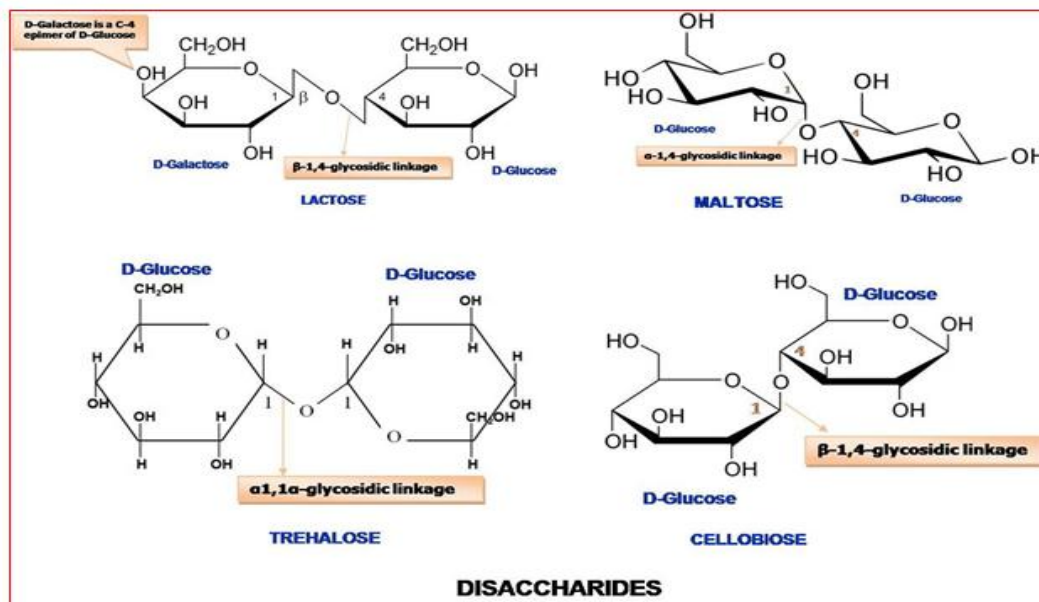
❖ Maltose :-

- **Systemic name:** 4-O- α -D-glucopyranosyl-D-glucose
- **Abbreviated name:** Glc($\alpha 1\rightarrow 4$)Glc
- **Molecular formula:** C₁₂H₂₂O₁₁
- **Common name:** Malt sugar
- **Structure:**
 - Maltose is a disaccharide produced through a dehydration or condensation process between two glucose molecules as monosaccharide unit.
 - Two D-glucose monomers are linked through a glycosidic bond between the α -anomeric carbon of C-1 of one glucose and the hydroxyl (-OH) oxygen atom on C-4 of the another glucose. This linkage is named as α -1,4-glycosidic bond hence the abbreviated name is Glc($\alpha 1\rightarrow 4$)Glc.

- Since the disaccharide have a free anomeric carbon of C-1 in the glucose, the maltose is a reducing disaccharides.
- Maltose is the second component of a major biochemical sequence of glucose chains. The addition of third glucose unit produces maltotriose; further additions will form dextrans (also named maltodextrins) and finally will form starch.

➤ **Importance:**

- Maltose is hydrolyzed rapidly into two monomer units of glucose by the action of maltase in living organisms. But, in the laboratory, heating maltose solution with a strong acid for more than a few minutes produces two glucose molecules.
- Maltose produced from germinating cereals, for example barley, which is an essential step for the brewing process. Germinating barley is malted in the process of brewing process in which maltose producing activity of amylases are maximized.
- Another process named as mashing in which these amylases breaks the cereal's starches into maltose. During fermentation, yeast metabolizes the maltose eventually produces ethanol and carbon dioxide.



Structure of important disaccharides i.e. lactose, maltose, trehalose and cellobiose.

❖ Trehalose :-

- **Systemic name:** α -D-glucopyranosyl- α -D-glucopyranoside
- **Abbreviated name:** Glc(α 1 $\leftrightarrow$ 1 α)Glc
- **Molecular formula:** C₁₂H₂₂O₁₁
- **Common name:** Mushroom sugar or mycose or tremalose
- **Structure:**
 - Trehalose is a naturally occurring alpha-linked disaccharide produced by glycosidic linkage between α - anomeric carbon at C-1 one D-glucose and α - anomeric carbon at C-1 of another D-glucose unit
 - Like sucrose, trehalose is also a non-reducing disaccharide.
- **Importance:**
 - The trehalose is broadly present in a number of organisms such as bacteria, fungi, plants, invertebrates and mammals.
 - Trehalose is a main component of the circulating fluid (hemolymph) in insects and is an energy-storage disaccharide. The enzyme trehalase which is a glycoside hydrolase, rapidly hydrolyzes trehalose into two glucose units. The constituent glucose unit is then readily absorbed in the gut and extensively used by insect for flight.
 - Trehalose is also found in plants such as sunflower seeds, *Selaginella* mosses, and sea algae.
 - Trehalose is also abundant in the various species of fungus. Fungi, for example, mushrooms shiitake (*Lentinula edodes*), maitake (*Grifola fondosa*) and nameko (*Pholiota nameko*) constitute 1-17% of trehalose by dry weight. Trehalose is too present in baker's yeast and wine yeast.
 - Trehalose has vital implication in the natural phenomena of anhydrobiosis. To withstand desiccation, many plants and animals invertebrates enter in the state of anhydrobiosis thus known as anhydrobiotic organisms. Anhydrobiosis requires accretion of compatible osmolytes, like the non-reducing disaccharide trehalose in order to protect from dehydration damage.

- Trehalose has tendency to retain high water content and thus used in food and cosmetics. This disaccharide sugar creates a gel phase when cells dehydrate that protects from disruption of internal cell organelles through efficiently supporting them in position.

❖ Cellobiose :-

- **Systemic name:** 4-O- β -D-glucopyranosyl-D-glucose
- **Abbreviated name:** Glc(β 1 $\rightarrow$ 4)Glc
- **Molecular formula:** C₁₂H₂₂O₁₁
- **Structure:**
 - Cellobiose is a disaccharide produced from the condensation of two units of same monosaccharide D-glucose joined through a β (1 $\rightarrow$ 4) glycosidic linkage. It is broken into two glucose units the action of bacteria or cationic ion exchange resins.
 - Cellobiose produced also from the hydrolysis of cellulose that has two D-glucose subunits.
 - Cellobiose different from maltose since *maltose* has α (1 $\rightarrow$ 4) glycosidic bond whereas *cellobiose* has β (1 $\rightarrow$ 4) glycosidic linkage between the two glucose subunits.
 - Both maltose and cellobiose exists in alpha and beta forms through mutarotation. This is because the hydroxyl (-OH) group linked to the anomeric carbon (of right glucose) not participating in an acetal formation may be in either an axial position or an equatorial position.
 - Like maltose, cellobiose is a reducing disaccharide since the one anomeric carbon of one glucose unit is free for oxidation (in other words it is hemiacetal).
- **Importance:**
 - Cellobiose may be produced through an enzymatic or acidic hydrolysis of cellulose and cellulose rich products for instance cotton, jute and paper.

Oligosaccharides

Oligosaccharides are produced from condensation between three to ten monosaccharide units and joined by glycosidic linkages. Maltotriose is an example. In cells, most of the oligosaccharides containing three or more monosaccharide units do not occur as free entity instead linked to non sugar molecules such as lipids or proteins forming glycoconjugates. Few examples of oligosaccharides are discussed below:

❖ Raffinose :-

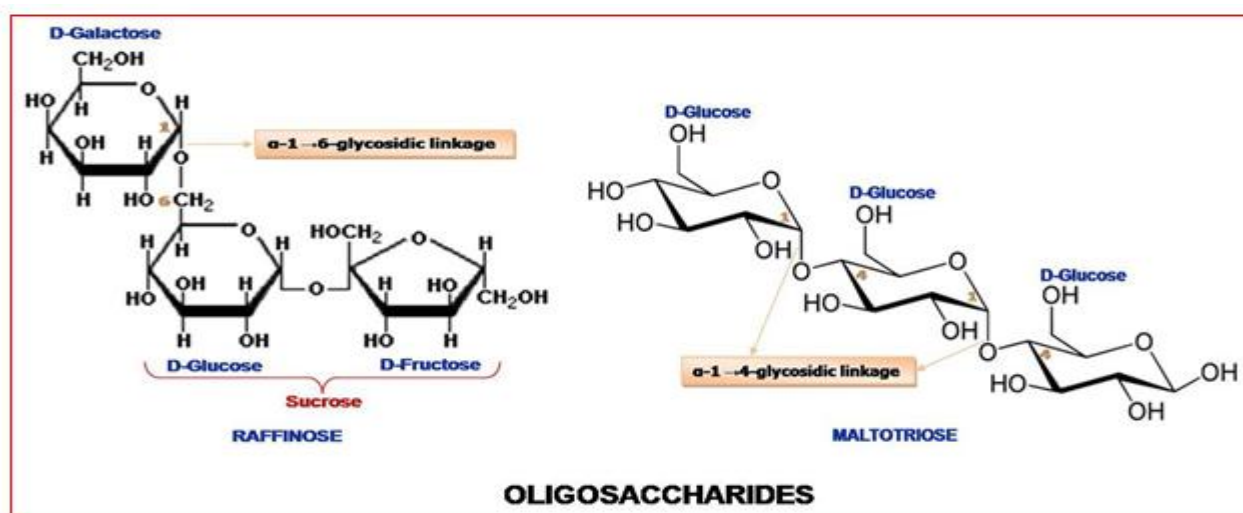
- **Systemic name:** β -D-Fructofuranosyl-O- α -D-galactopyranosyl-(1 $\rightarrow$ 6)- α -D- glucopyranoside
- **Molecular formula:** C₁₈H₃₂O₁₁
- **Structure:**
 - Raffinose, also known as melitose, formed from three monosaccharide units namely galactose, fructose, and glucose and thus is a trisaccharide
 - In raffinose, galactose attached to sucrose (glucose+fructose) through a 1 α $\rightarrow$ 6 glycosidic linkage.
- **Importance:**
 - Raffinose is mostly present in legumes and cruciferous vegetables such as beans, peas, cabbage, brussels sprouts, and broccoli. Raffinose may be hydrolyzed into D-galactose and sucrose through an enzyme α -galactosidase (called α -GAL in short hand notation).
 - Humans and other monogastric animals such as pigs and poultry do not have α -GAL enzyme to break down Raffinose. Thus these oligosaccharides remain undigested and pass through the stomach and upper intestine. In the lower intestine, these are fermented through gas-producing bacteria that have the α -GAL enzyme and produce carbon dioxide and methane which in turn creates flatulence.

❖ Maltotriose :-

- **Systemic name:** α -D-Glucopyranosyl-(1 $\rightarrow$ 4)- α -D-glucopyranosyl-(1 $\rightarrow$ 4)-D-glucose
- **Molecular formula:** C₁₈H₃₂O₁₆
- **Structure:**
 - Maltotriose is a trisaccharide that constitutes three glucose units attached through α -1,4 glycosidic linkages

➤ Importance:

- It is mostly formed through a digestive enzyme called alpha-amylase (a commonly found in human saliva) that works on amylose in starch. Production of both maltotriose and maltose in this process occurs due to the random manner in which alpha amylase cleaves α -1,4 glycosidic linkages.



Structure of oligosaccharides i.e. raffinose and maltotriose.

Polysaccharides

Most of the carbohydrates abundantly present in nature in the form of polysaccharides. Polysaccharides are very large bio-molecules and polymers of high molecular weight. Polysaccharides (poly= “many”) constituting ten or more monosaccharide units linked together through glycosidic bonds.

The monosaccharide chain in the polysaccharides can be branched or unbranched (linear), and monosaccharides present can be of same or different types. Thus polysaccharides, also known as **glycans**, vary largely in the types of their repeating monosaccharide units, in the chain length of their chains, in the types of glycosidic linkages attaching the monomer units, and in the degree of branching.

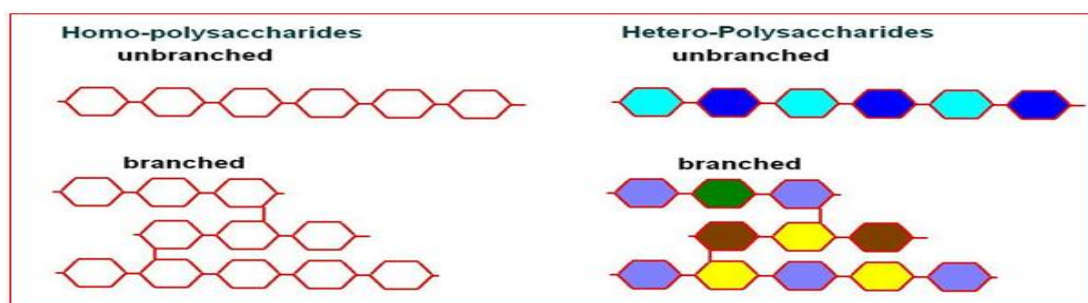
Polysaccharides play vital roles in energy storage upon hydrolysis into monosaccharide units specifically D-glucose and in retaining the structural integrity of an organism.

Types of polysaccharides

Basically, polysaccharide is categorized into homopolysaccharides and heteropolysaccharide depending on the variety of monosaccharide units attached in the sugar chain.

A. Homopolysaccharides may be branched or unbranched but they contain the same monosaccharide unit.

B. Heteropolysaccharides can also be branched or unbranched but they contain two or more types of monosaccharide unit. Heteropolysaccharides serve as an extracellular support in tissues of all organisms.



Branching pattern of monosaccharide units in homopolysaccharides and heteropolysaccharides. The different colors represent different types of monosaccharides.

A. Homopolysaccharides

Some homopolysaccharides such as starch and glycogen, provide storage forms of energy. Other homopolysaccharides such as cellulose and chitin constitute structural components of plant cell walls and animal exoskeletons. Different homopolysaccharides categorized according to their functions are discussed below:

I. Homopolysaccharides as Energy Storing Forms

The homopolysaccharides as energy storage form are starch in plants and glycogen in animals. Both polysaccharides occur within cells in large clusters or granules forms. Starch and glycogen molecules are greatly hydrated since they constitute many exposed hydroxyl (-OH) groups accessible for hydrogen-bonding with water. In addition, inulins are also energy storing polysaccharides found in plant cells.

Advantage of Glycogen and Starch over Glucose

Glucose is an important energy source in virtually all life forms. However, free glucose molecules cannot be stored because if glucose stored in high amounts, glucose will disturb the osmotic balance of cell, eventually results in cell death. The solution is to store glucose as units in large polymer, which is not osmotically active. Thus, glycogen and starch are movable stores of glucose.

❖ Starch:-

Starch is the stored form of sugars in plants and has two forms, *amylose* and *amylopectin* both of which are polymers of glucose.

➤ **Structure:**

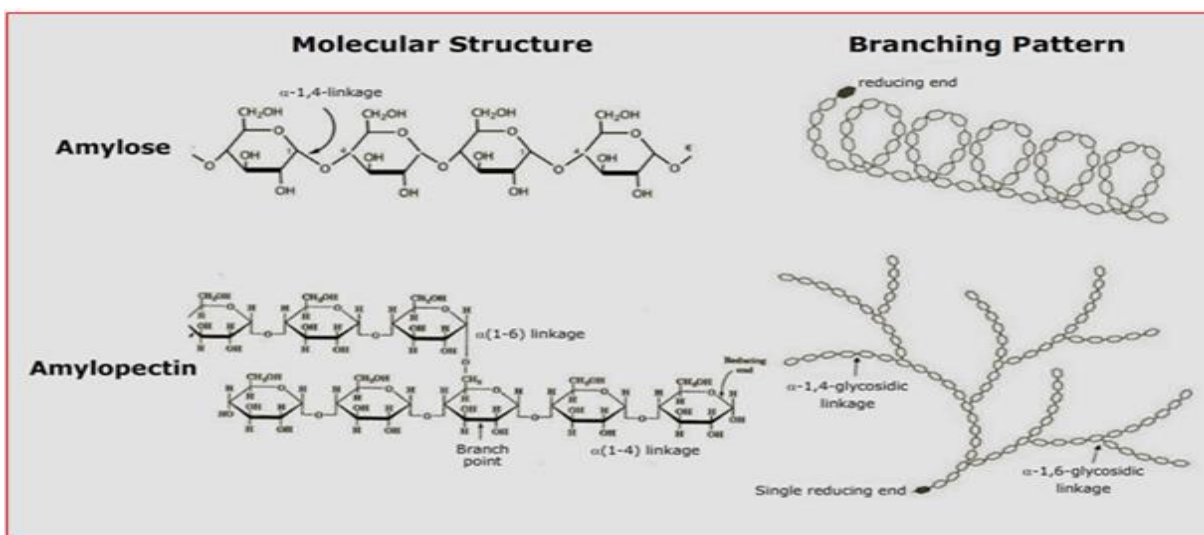
Starch constitutes glucose as monomers linked through α 1-4 or α 1-6 glycosidic bonds.

- ❖ Amylose,: the long unbranched (linear) type of starch, contains glucose as monomers linked through α -1,4 glycosidic linkages (as in maltose). These glucose chains vary in molecular weight from a few thousand to more than a million. Amylase makes up approximately 20-30% of the structural components of starch. Due to hydrogen bonding, amylose acquires a coiled or helical structure which comprised of six glucose units per turn.

When starch mixed with iodine solution the amylose-iodine complex is formed that produce distinct blue-violet colour. This occurs as coiled structure of amylose has sufficient space in its core to accommodate iodine molecules.

- ❖ **Amylopectin**, the branched form of starch that contains about one α -1,6 glycosidic linkage per 24-30 α -1,4 glycosidic linkages, in similar pattern to glycogen excluding lower degree of branching in amylopectin. Amylopectin has a high molecular weight up to 200 million. Amylopectin makes up 70–80% of the structure of starch.

The helical structure of amylopectin is interrupted through the branching of the glucose chain, therefore instead of the deep blue-violet color which amylose gives with iodine, amylopectin generates a reddish brown colour.



Structure and branching pattern of Amylose and Amylopectin

➤ Importance:

- Plants can synthesize glucose and the excess glucose is stored as starch in different plant parts such as roots and seeds.
- Most plant cells have the ability to form starch and starch storage is particularly abundant in tubers (underground stems), such as potatoes and in seeds.
- The cells can then absorb the glucose. More than half the carbohydrate ingested by human beings is starch found in wheat, potatoes, and rice and so on. The starch that is consumed by animals is broken down into glucose.

❖ Glycogen :-

Glycogen is the homopolysaccharide abundantly found in animal cells and it serve as the storage form of glucose. Thus, glycogen in the animal is equivalent to starch in the plant.

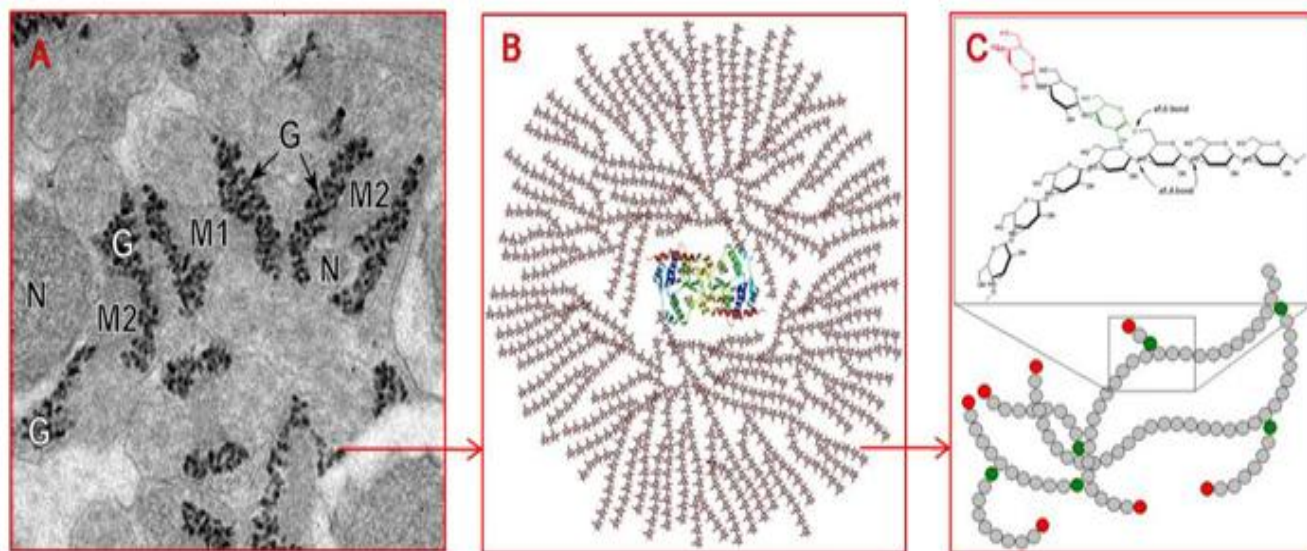
➤ **Structure:**

- Glycogen is a very large and highly branched polymer of glucose subunits.
- Like amylopectin in starch, glycogen is a homopolymer of (α 1,4)-linked subunits of linear glucose chain(**amylase**) containing (α 1,6)-connected branches of glucose (**amylopectin**). However, glycogen is more extensively branched that occurs after every 8 to 12 subunits of glucose in chain and more compact than starch.
- Each branch in glycogen ends with a non-reducing glucose unit. Therefore, single glycogen having n branches has $n+1$ non-reducing ends with only one reducing end
- When glycogen is needed for energy source, glucose molecules are removed one at a time from the non-reducing ends of glycogen. Enzymes responsible for degradation of glycogen which work only at non-reducing ends can work at the same time on the many branches thereby accelerating the conversion of the polymer to monosaccharides.

Importance:

- Glycogen is particularly stored in the cells of liver in which it constitute about 7% of the wet weight where it is synthesized in presence of excess free glucose as well as degraded when glucose levels get lower. Glycogen in skeletal muscle serves as an instant reserve source of glucose for muscle cells.
- In liver cells, glycogen occurs as large granules that are further clusters of smaller granules Each small granule constitute single and highly branched glycogen molecule with an average molecular weight up to several million.
- For glycogen synthesis in liver, insulin acts on the hepatocytes to activate a number of enzymes such as glycogen synthase. Eventually, the excess glucose molecules available in the body are added to the chains of glycogen for storage.
- When glucose level decreases in the body, many enzymes such as glycogen phosphorylase in liver catalyze the hydrolysis of glycogen in to its monomer unit glucose and transported through blood stream.

- Glycogen and starch containing both the forms amylopectin, amylose are rapidly hydrolyzed by breaking the alpha (α 1 $\rightarrow$ 4) glycosidic linkage through the action of α -amylase secreted by the salivary glands and the pancreas.

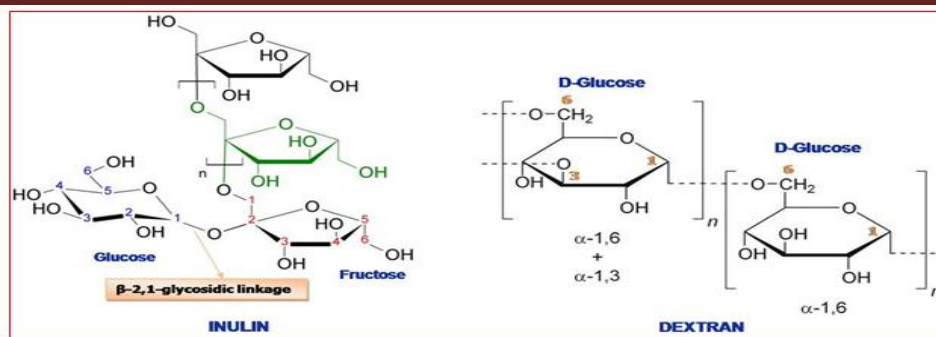


A. Glycogen as small granules in clusters. **B.** Schematic 2-D cross-sectional view of glycogen where a core protein of glycogenin is flanked by branches of glucose units. A single globular granule can have about 30,000 glucose units. **C.** Molecular structure of small section of glycogen where α 1 $\rightarrow$ 4-linked subunits of linear glucose chain contains (α 1 $\rightarrow$ 6)-connected (in green) branches of glucose with non-reducing ends (in red).

❖ Inulins:-

Inulins are naturally occurring homopolysaccharides synthesized in various types of plants, very often obtained from chicory for industrial purposes.

- **Structure:** Inulin are a polymer chain of fructose molecules which ends with a glucose molecule. In inulins, glucose and fructose joined through β (2,1) linkage.
- **Importance:**
 - The inulins classified in to dietary fibers and called as fructans.
 - Inulin is utilized in some plants as a storing energy form instead of starch and is generally abundant in roots and rhizomes in more than 36,000 species of plants such as wheat, onion, bananas, garlic, asparagus, artichokes and chicory.



Structural formula of inulin and dextran. In inulin, additional features (carbon numbers and repeated fructosyl unit) are highlighted with colours where $n \sim 35$.

❖ Dextrans:-

Dextrans, a homopolysaccharides commonly found in bacterial and yeast.

- **Structure:** Dextrans constitute polymer of (α 1 $\rightarrow$ 6)-linked linear chain of glucose with (α 1 $\rightarrow$ 3) linked branches of glucose, and a few also contain (α 1 $\rightarrow$ 2) or (α 1 $\rightarrow$ 4) branches.
- **Importance:**
 - Bacteria growing on the surface of teeth forms plaque that is rich in dextrans.
 - Artificial dextrans are utilized in many commercial products such as sephadex which serve in the fractionation of proteins through size-exclusion chromatography.
 - The dextrans in commercial products are chemically cross-linked to constitute insoluble materials of different porosity admissible to varying sizes of macromolecules.

II. Homopolysaccharides as structural forms

❖ Cellulose:-

Cellulose, a homopolysaccharide, is fibrous, compact and water-insoluble carbohydrate. It is abundant in the cell walls of plants, especially in stalks, stems, trunks and all the woody parts of the plant.

- **Structure:**
 - Cellulose is a homopolymer of glucose molecules attached through β -1,4 glycosidic linkages, in contrast with α -1,4 linkages seen in starch and glycogen.

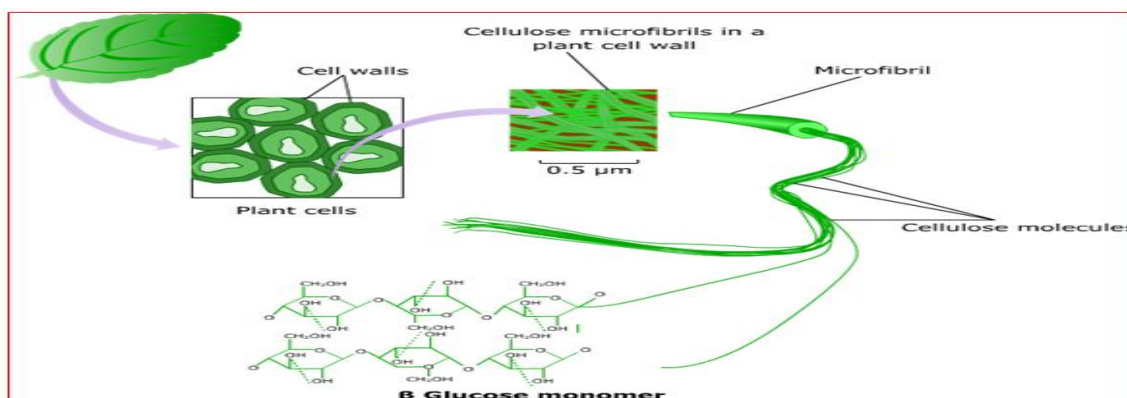
- Like amylose, the cellulose molecule is a linear and unbranched homopolymer containing around 10,000 to 15,000 D-glucose molecules.

➤ **Importance:**

- Cellulose is the most abundant polysaccharide on the earth. About 10^{15} kg of cellulose is synthesized and degraded per year which is an amount 1000 times as great as the combined weight of the human race.
- The cellulose constitutes major proportion of plant cell walls that serves structural support to the cell. Wood, cotton and paper are mostly cellulosic in nature. Cellulose passing through our digestive organ is termed as dietary fiber.
- As shown in, alternate glucose monomer in cellulose is turned over, and the monomers are compact throughout extended long chains. This arrangement of glucose gives cellulose its stiffness and high tensile strength, in this way it serves structural support to plant cells.

➤ **Role of β configuration in cellulose as structural component:**

- The β configuration of glucose in cellulose is to constitute longer and linear chains. Consequently, parallel chains of glucose interconnect with one another by hydrogen bonds called as microfibrils of cellulose.
- The α -1,4 bonds in starch and glycogen create a very different molecular construction to that in cellulose. A hollow helix is formed in glycogen and starch instead of a straight chain.
- These distinct upshots of α and β glycosidic linkages have biological significance. The straight chain produced through beta linkages is most favorable for the construction of fibers with a high tensile strength. On the contrary, the open helix produced by alpha linkages is favorable to produce a storage form of glucose.



In cellulose, glucose monomers in linear chains are attached through β 1-4 glycosidic bonds. Every glucose monomer is flipped relative to the next one produces a linear and fibrous structure.

➤ Cellulose digestion:

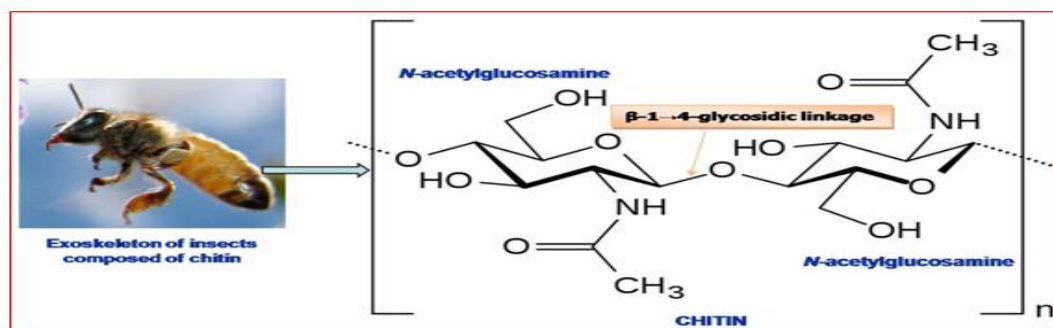
- Most animals including human being cannot use cellulose as an energy source, since they do not have an enzyme to hydrolyze the (β 1->4) glycosidic linkages unlike (α 1->4) linkages in glycogen and starch that is broken to glucose unit with the help of α -amylases enzyme.
- Herbivores, for instance, cows, buffalos and horses have some species of bacteria and protists that inhabit in the rumen (portion of the digestive system) and produce the enzyme cellulase. Cellulases catalyze the hydrolysis of (β 1->4) linkages in cellulose thus produce glucose monomer units to be used as an energy source by these animals.
- Termites can also digest cellulose and thus wood, since their intestinal tract has a symbiotic microorganism, *Trichonympha* which produce cellulase.

❖ Chitin:-

Arthropods, such as insects, spiders, and crabs, possess an outer skeleton known as exoskeleton that guards their internal body parts. This exoskeleton is made of the bio-molecule termed as chitin.

➤ Structure:

- Chitin is a linear homopolysaccharide comprised of repeating units of *N*-acetylglucosamine, modified monosaccharide containing nitrogen and joined through β 1->4 glycosidic linkages.
- The only difference in structure of chitin and cellulose is that the hydroxyl group at C-2 (in cellulose) replaced with an acetylated amino group (in chitin).
- Like cellulose, chitin also constitutes extended fibers and cannot be digested by vertebrates.



Structure of Chitin.

➤ **Importance:**

- Chitin is the chief constituent of the hard exoskeletons of about million species of arthropods.
- It might be the second most abundant carbohydrate next to cellulose in biosphere as about 1 billion tons of chitin are produced each year in the biosphere.
- Chitin is also a main constituent of cell walls of fungus.

B. Heteropolysaccharides

In general, heteropolysaccharides (heteroglycans) contain two or more different monosaccharide units. Although a few representatives contain three or more different monosaccharides, most naturally occurring heteroglycans contain only two different ones and are closely associated with lipid or protein. The complex nature of these substances has made detailed structural studies extremely difficult. The major heteropolysaccharides include the connective-tissue polysaccharides, the blood-group substances, glycoproteins (combinations of carbohydrates and proteins) such as gamma globulin, and glycolipids (combinations of carbohydrates and lipids), particularly those found in the central nervous system of animals and in a wide variety of plant gums.

Representative Heteropolysaccharides

Heteropolysaccharide	Component sugars	Functions	Distribution
Hyaluronic acid	D-glucuronic acid and <i>N</i> -acetyl-D-glucosamine	lubricant, shock absorber, binding water	connective tissue, skin
Chondroitin-4-sulfate*	D-glucuronic acid and <i>N</i> -acetyl-D-galactosamine-4-O-sulfate	Calcium accumulation, cartilage and bone formation	cartilage
Heparin*	D-glucuronic acid, <i>L</i> -iduronic acid, <i>N</i> -sulfo-D-glucosamine	anticoagulant	mast cells, blood
Gamma globulin*	<i>N</i> -acetyl-hexosamine, D-mannose, D-galactose	antibody	blood
Blood group substance*	D-glucosamine, D-galactosamine, <i>L</i> -fucose, D-galactose	blood group specificity	cell surfaces, especially red blood cells
*Covalently linked to protein; the proportion of protein to carbohydrate in such complex molecules varies from about 10% protein in the case of chondroitin-4-sulfate to better than 95% for gamma globulin.			

The most important heteropolysaccharides are found in the connective tissues of all animals and include a group of large molecules that vary in size, shape, and interaction with other body substances. They have a

structural role, and the structures of individual connective-tissue polysaccharides are related to specific animal functions; hyaluronic acid, for example, the major component of joint fluid in animals, functions as a lubricating agent and shock absorber.

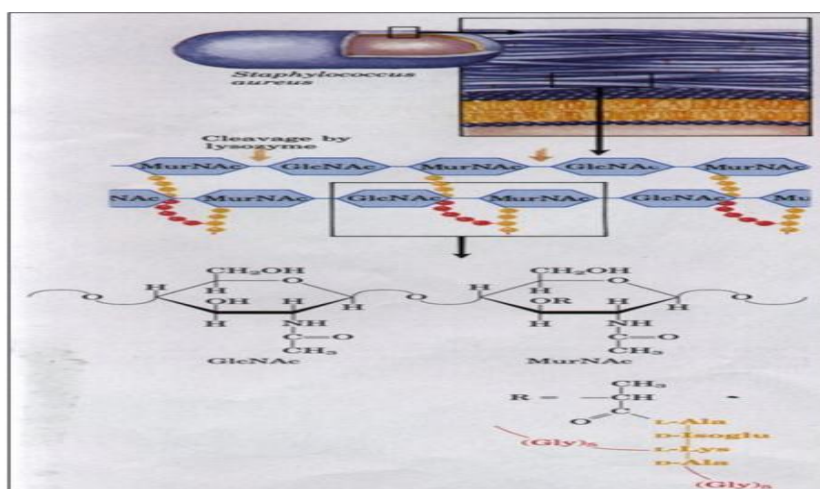
The connective-tissue heteropolysaccharides contain acidic groups (uronic acids or sulfate groups) and can bind both water and inorganic metal ions. They can also play a role in other physiological functions; e.g., in the accumulation of calcium before bone formation. Ion-binding ability also appears to be related to the anticoagulant activity of the heteropolysaccharide heparin.

The size of the carbohydrate portion of glycoproteins such as gamma globulin or hen-egg albumin is usually between five and 10 monosaccharide units; several such units occur in some glycoprotein molecules. The function of the carbohydrate component has not yet been established except for glycoproteins associated with cell surfaces; in this case, they appear to act as antigenic determinants (i.e., they are capable of inducing the formation of specific antibodies).

Many types of heteropolysaccharides occurs some of which found in bacterial and algal cell walls and some constitute an extracellular matrix in animal tissues that holds each cell together, gives protection and maintains shape of cell. Different heteropolysaccharides categorized corresponding to their role in organisms are as follows:

✚ Heteropolysaccharides in Bacterial and Algal Cell Walls

❖ Peptidoglycan: Constituents of bacterial cell walls :-



The peptidoglycan of the cell wall of the gram-positive bacterium *Staphylococcus aureus*. Peptides (red and yellow) attached to N-acetylmuramic acid units in two neighboring chains covalently link the polymers. Note the mixture of L and D amino acids in the peptides. Isoglu represents isoglutamate, in which the side chain carboxyl group, not the carboxyl at C-1, is involved in a peptide bond.

➤ **Structure:**

- It is a heteropolysaccharide comprised of alternating (β 1- $\rightarrow$ 4)-linked *N*-acetylglucosamine and *N*-acetylmuramic acid moieties.
- Peptidoglycan contains lactic acid (a three carbon carboxylic acid) that is ether-linked to the oxygen at C3 of *N*-acetylglucosamine.

➤ **Importance:**

- Peptidoglycan is the rigid constituent of bacterial cell walls.
- The enzyme lysozyme destroys bacteria through breaking the (β 1- $\rightarrow$ 4) glycosidic linkage between *N*-acetylglucosamine and *N*-acetylmuramic acid of peptidoglycan. Lysozyme is particularly found in tears, as it defense against bacterial infections of the eye.

❖ **Agar: Constituents of algal cell walls :-**

➤ **Structure:**

- Agar is a mixture of sulphated heteropolysaccharides in which D-galactose and an L-galactose are attached through ether-linked between C-3 and C-6.
- Agarose is the agar component containing few charged groups such as sulfates, pyruvates. agarose is useful in the biochemical experiments because of its significant gel-forming property.

➤ **Importance:**

- Agar is the chief constituent of cell walls in some marine red algae such as seaweeds.

 Heteropolysaccharides in Extracellular Matrix

The extracellular matrix (ECM), also called ground substance which encloses fibroblasts and other connective tissue cells is composed of an interlocking meshwork of heteropolysaccharides and fibrous proteins, for instance, fibrillar collagens, elastin, and fibronectin. Specialized ECM which lies beneath epithelial cells constitute basement membrane and it made up of specialized collagens, laminin and heteropolysaccharides. The variety of heteropolysaccharides that constitute the ECM basically called glycosaminoglycans. Structure and function of glycosaminoglycans and their types are explained below:

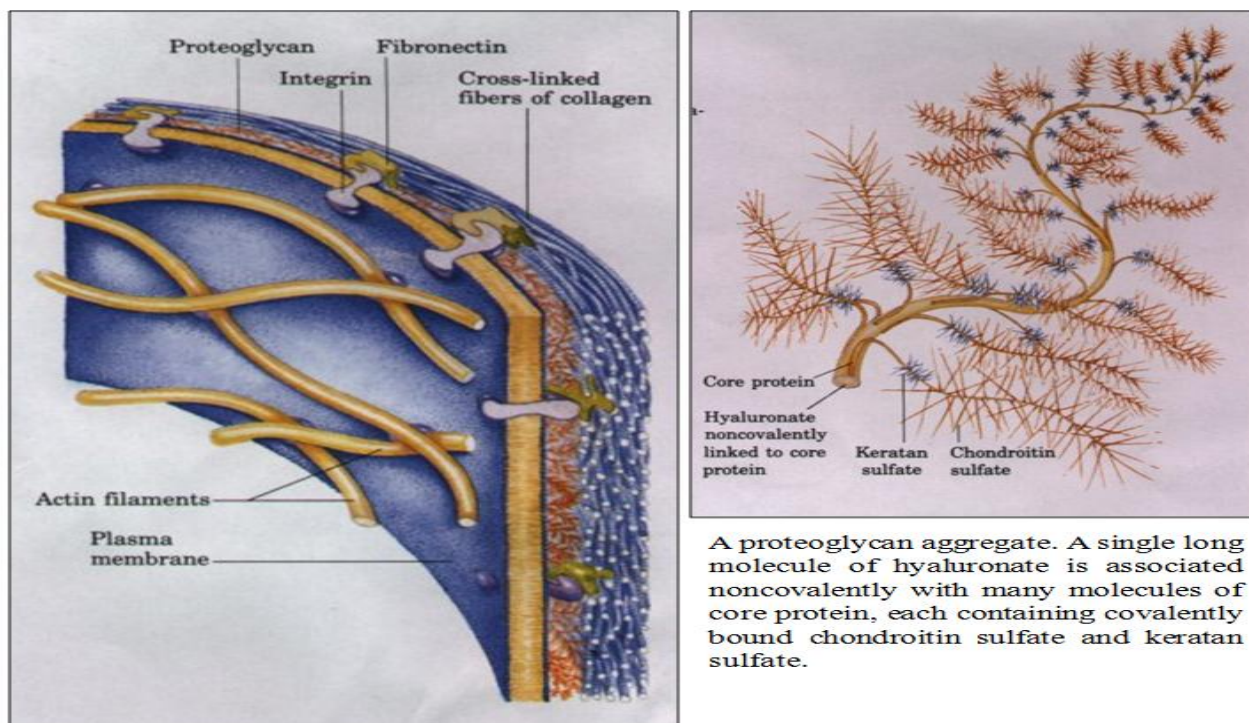
The extracellular space in animal tissues is filled with a gel-like material, the extracellular matrix, also called ground substance, which holds the cells of a tissue together and provides a porous pathway for the diffusion of nutrients and oxygen to individual cells. The extracellular matrix is composed of an interlocking meshwork of heteropolysaccharides and fibrous proteins.

The heteropolysaccharides, called glycosaminoglycans, are a family of linear polymers composed of repeating disaccharide units. One of the two monosaccharides is always either N-acetylglucosamine or N-acetylgalactosamine; the other is in most cases a uronic acid, usually glucuronic acid. In some glycosaminoglycans, one or more of the hydroxyls of the amino sugar is esterified with sulfate. The combination of these sulfate groups and the carboxylate groups of the uronic acid residues gives the glycosaminoglycans a very high density of negative charge. To minimize the repulsive forces among neighboring charged groups, these molecules assume an extended conformation in solution. One consequence of this extended conformation is the very high viscosity of solutions of these long, thin molecules. Glycosaminoglycans are attached to extracellular proteins to form proteoglycans (discussed below), enormous aggregates in which the polysaccharide makes up most of the mass, often 95% or more.

The glycosaminoglycan hyaluronic acid (hyaluronate at physiological pH) of the extracellular matrix of animal tissues contains alternating units of D-glucuronic acid and N-acetylglucosamine. Hyaluronates have molecular weights greater than 1 million; they form clear, highly viscous solutions, which serve as lubricants in the synovial fluid of joints, and give the vitreous humor of the vertebrate eye its jellylike consistency. Hyaluronate is also a central component of the extracellular matrix of cartilage and tendons, to which it contributes tensile strength and elasticity. Hyaluronidase, an enzyme secreted by some pathogenic (disease-causing) bacteria, can hydrolyze the glycosidic linkages of hyaluronate, rendering tissues more susceptible to invasion by the bacteria. A similar enzyme in sperm hydrolyzes an outer glycosaminoglycan coat around the ovum of many organisms, allowing sperm penetration.

Proteoglycans are composed of a very long strand of hyaluronate to which numerous molecules of core protein are bound noncovalently, at about 40 nm intervals. Each core protein is bound covalently to many shorter glycosaminoglycan molecules, such as chondroitin sulfate, keratan sulfate, heparan sulfate, and dermatan sulfate. The covalent attachments between glycosaminoglycans and core protein are glycosidic bonds between sugar residues and the hydroxyl groups of Ser residues in the protein. A typical proteoglycan in human cartilage contains about 150 polysaccharide chains (each of $M_r \approx 20,000$) covalently bound as side chains to each core protein. When a hundred or more of these "decorated" core

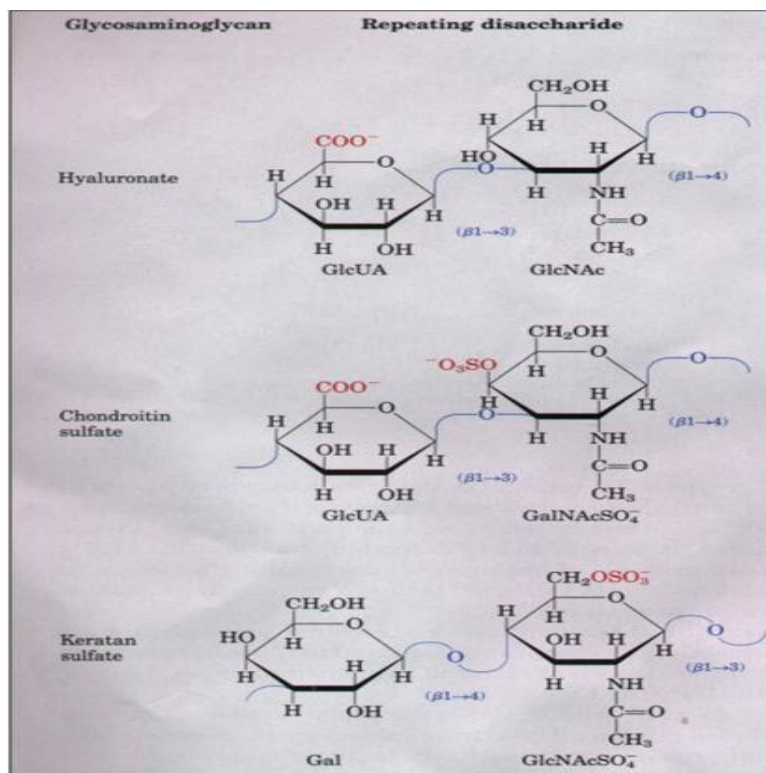
proteins bind a single, extended molecule of hyaluronate, the resulting proteoglycan and its associated water of hydration occupy a volume about equal to that of an entire bacterial cell.



The association between cells and the proteoglycan of extracellular matrix is mediated by a membrane protein (integrin) and an extracellular protein (fibronectin in this example) with binding sites for both integrin and the proteoglycan. Note the close association of collagen fibers with the fibronectin and proteoglycan.

Interwoven with these enormous extracellular proteoglycans are fibrous proteins such as collagen and elastin, which form a cross-linked meshwork that gives the whole extracellular matrix strength and resiliency.

The attachment of cells to the extracellular meshwork involves several families of proteins. The extracellular domains of certain integral membrane proteins (integrins) have binding sites for another family of adhesion proteins (including fibronectin and laminin), that bind to proteoglycans. It is likely that this complex system of binding proteins serves not merely to anchor cells to the extracellular matrix, but also to direct the migration of cells in developing tissue along paths determined by the organization of the extracellular matrix.

❖ Glycosaminoglycans:

Some of the common heteropolysaccharide (glycosaminoglycan) components of extracellular matrix. The ionized carboxylate and sulfate groups give these polymers their characteristic high negative charge.

➤ **Structure:**

- The glycosaminoglycans are a family of unbranched heteropolysaccharides that contains repeating disaccharide units .
- In a disaccharide unit of glycosaminoglycans, one monosaccharide is always either *N*-acetylglucosamine or *N*-acetylgalactosamine and the second monosaccharide in most of the glycosaminoglycans is an uronic acid, generally D-glucuronic or L-iduronic acid.
- A few glycosaminoglycans contain esterified sulfate groups. When sulfate groups are combined with the carboxylate groups of the uronic acid moieties, it makes glycosaminoglycans with a very high density of negative charge.

➤ **Importance:**

- Glycosaminoglycans are exclusive in animals and bacteria and are not found in plants.
- The sulfated glycosaminoglycans are attached to extracellular proteins to form proteoglycans.

➤ Types of Glycosaminoglycans:

Major types of glycosaminoglycans include hyaluronan, chondroitin sulfate, keratan sulfate, heparan sulfate and dermatan sulfate. Glycosaminoglycans such as chondroitin sulfate, keratan sulfate, heparan sulfate and dermatan sulfate are different from hyaluronan in three aspects: (1) they are usually much shorter polymers; (2) they are covalently attached to specific proteins to constitute proteoglycans, and (3) one or both monomers in disaccharide unit different from those of hyaluronan.

1. Hyaluronan (hyaluronic acid):

✓ Structure:

- Hyaluronan (hyaluronic acid) comprised of alternating moieties of D-glucuronic acid and *N*-acetylglucosamine.
- As it contains up to 50, 000 repeats of the basic disaccharide unit, hyaluronan has a molecular weight of several million.

✓ Significance:

- It forms highly viscous substance and have a glassy or translucent appearance hence the name hyaluronan which is the Greek *hyalos* means glass.
- It serves as lubricants in the synovial fluid of joints and contributes jellylike consistency to the vitreous humor of the vertebrate eye.
- Hyaluronan is also a constituent of the extracellular matrix of cartilage and tendons where it provides tensile strength and elasticity due to which it has strong interactions with other components of the matrix.
- An enzyme called hyaluronidase produced in a few pathogenic bacteria that break the glycosidic bonds of hyaluronan, making tissues more sensitive to bacterial attack.
- A similar enzyme in sperm of several species hydrolyzes an outer glycosaminoglycan coat around the ovum, to let sperm penetrate.

2. Chondroitin sulfate:

✓ Structure:

- Chondroitin sulfate contains an alternating copolymer β -glucuronic acid-(1-3)-*N*-acetyl- β -galactosamine-4-sulfate.

✓ **Significance:**

- Chondroitin sulfate (the Greek *chondros*, means "cartilage") gives tensile strength to cartilage, tendons, ligaments and the walls of the aorta.

3. Dermatan sulfate:

✓ **Structure:**

- In this heteropolysaccharide, several glucuronate moieties present in chondroitin sulfate are replaced by their 5-epimer called L-iduronate.

✓ **Significance:**

- Dermatan sulfate (the Greek *derma* means "skin") provides to the pliability of skin and is also found in blood vessels and heart valves.

4. Heparan sulfate:

✓ **Structure:**

- Heparin contains primarily iduronic acid (IdoA) and a smaller proportion of glucuronic acid (GlcA).

✓ **Significance:**

- Heparin is a fractionated form of heparan sulfate derived mostly from mast cells (a type of leukocyte).
- Heparin is a therapeutic agent used to prevent blood coagulation as it has ability to bind the protease inhibitor antithrombin. Heparin binding causes antithrombin to bind to and inhibit thrombin, a protease essential to blood clotting.
- Purified heparin is regularly added in blood samples taken for clinical examination and to blood donated for transfusion to inhibit clotting.

❖ Glycoconjugates :-

Besides, carbohydrate's role as energy storage and structural materials, carbohydrates (polysaccharides and oligosaccharides) play significant roles as information carriers. In order to act as information carrier, many of the carbohydrates are covalently attached to a protein or a lipid to constitute glycoconjugates.

✓ Functions of glycoconjugates

- Some of them provide communication between cells and their extracellular surroundings. Others label proteins for transport to and localization in specific organelles or for destruction when the protein is malformed or superfluous.
- Others serve as recognition sites for extracellular signal molecules (growth factors, for example) or extracellular parasites (bacteria or viruses).
- On almost every eukaryotic cell, specific oligosaccharide chains attached to components of the plasma membrane form a carbohydrate layer (the glycocalyx), several nanometers thick, that serves as an information-rich surface that a cell shows to its surroundings.
- These oligosaccharides have central roles in cell-cell recognition and adhesion, cell movement during development, blood clotting, immune response, wound healing and other cellular processes.

✓ Classifications of glycoconjugates

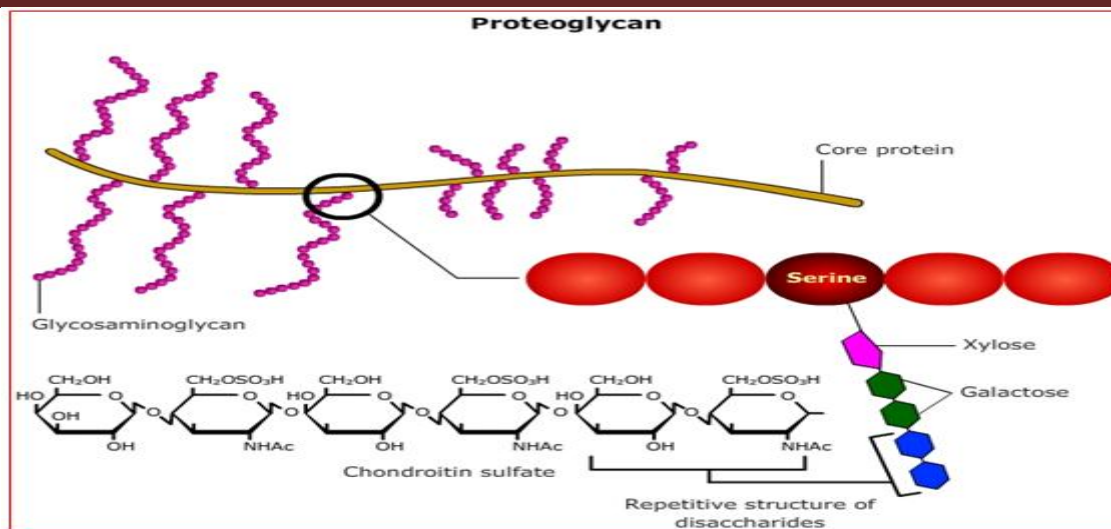
Three major classes of glycoconjugates are Proteoglycans, Glycoproteins, and Glycolipids

1. Proteoglycans

Proteoglycans resemble more to polysaccharides than to proteins as it contains 95% carbohydrate moieties by weight.

➤ Structure:

- Proteoglycans are structural constituent of extracellular matrix in which one or more sulfated glycosaminoglycan chains are attached covalently to a membrane protein or a secreted protein termed as "core protein".
- The point of attachment is a Serine (Ser) residue, to which the glycosaminoglycan such as chondroitin sulfate is attached through a trisaccharide bridge i.e. Gal-Gal-Xyl
- The Serine residue is usually in the sequence -Ser-Gly-X-Gly- (where X is any amino acid residue), although not every protein with this sequence has an attached glycosaminoglycan.



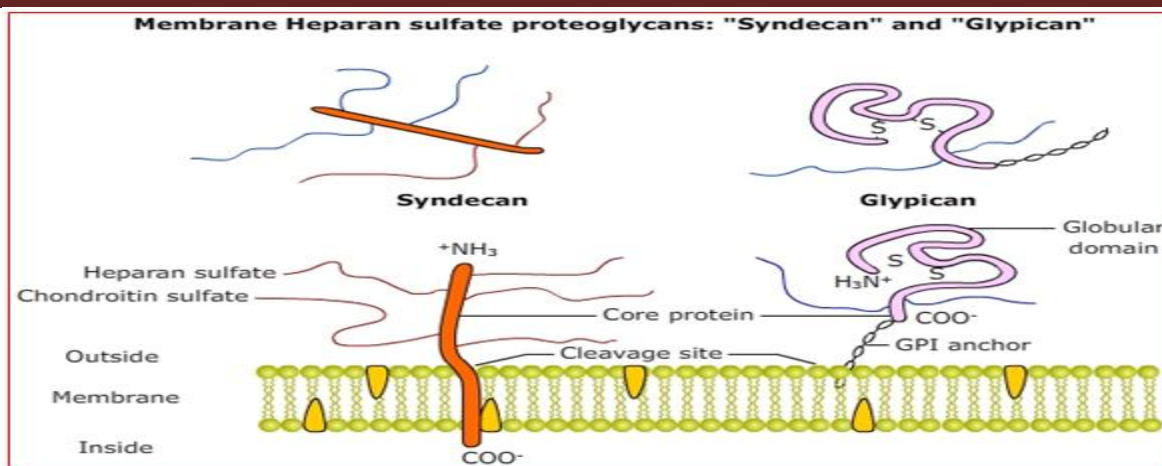
Proteoglycan structure, showing the trisaccharide bridge.

➤ **Importance:**

- Proteoglycans are major components of all extracellular matrices.
- Proteoglycans work as lubricants and structural components in connective tissue, mediate adhesion of cells to the extracellular matrix and bind factors that stimulate cell proliferation.
- Mammalian cells can synthesize 40 types of proteoglycans. These molecules act as tissue organizers and they influence various cellular activities, for instance, growth factor activation and adhesion.

➤ **Types of Proteoglycans:**

There are two major families of membrane heparan sulfate proteoglycans. **Syndecans** have a single trans-membrane domain and an extracellular domain bearing three to five chains of heparin sulfate and in some cases chondroitin sulfate. **Glypicans** are linked to the membrane through a lipid anchor, a derivative of the membrane lipid phosphatidylinositol. Both syndecans and glypicans may be shed into the extracellular space. A protease in the ECM that cuts close to the membrane surface liberates syndecan ectodomains (those domains outside the plasma membrane) and a phospholipase which breaks the connection to the membrane lipid releases glypicans. Many chondroitin sulfate and dermatan sulfate proteoglycans also exist, some as membrane-bound entities, others as secreted products in the ECM.



Two major families of membrane heparan sulfate proteoglycans. Schematic diagram of a syndecan and a glypican in the plasma membrane. Syndecans are held in the membrane by hydrophobic interactions between a sequence of non polar amino acid residues and plasma membrane lipids they can be released by a single proteolytic cut near the membrane surface. Glypicans are held in the membrane by a covalently attached membrane lipid GPI-anchor (glycosylphosphatidylinositol-GPI).

2. Glycoproteins:

➤ Structure:

- Glycoproteins constitute one or more oligosaccharides that attached covalently to a protein. Thus, glycoproteins are carbohydrate-protein conjugates.
- Glycans are smaller, branched and more structurally diverse oligosaccharides than the glycosaminoglycans of proteoglycans.
- About half of all proteins of mammals are glycosylated, and about 1% of all mammalian genes encode enzymes involved in the synthesis and attachment of these oligosaccharide chains.

➤ Significance:

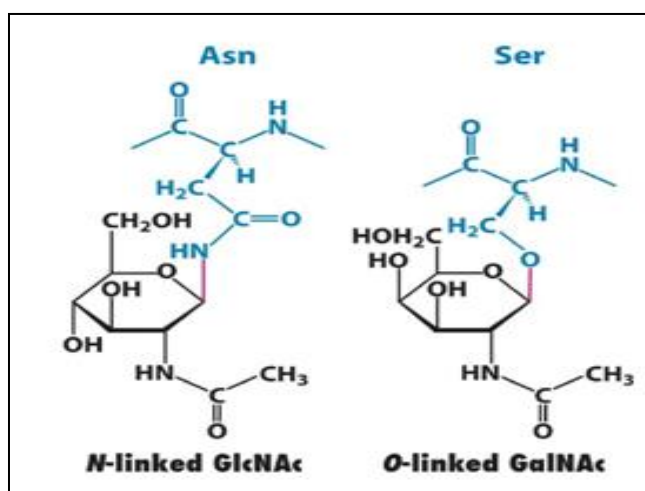
- Glycoproteins are generally present on the outer surface of the plasma membrane (as part of the glycocalyx), in the extracellular matrix and in the blood.
- Inside cells, these are particularly present in organelle e.g. golgi complexes, lysosomes and secretory granules. A few cytosolic and nuclear proteins can also be glycosylated.
- Like glycosaminoglycans, the oligosaccharide moieties of glycoproteins are rich in information, forming highly specific sites for recognition and high affinity binding by carbohydrate-binding proteins called lectins.
- Many of the proteins secreted by eukaryotic cells are glycoproteins, such as most of the proteins of blood. Immunoglobulins (antibodies) and some hormones, like follicle-stimulating

hormone, luteinizing hormone, and thyroid-stimulating hormone are glycoproteins. Several milk proteins e.g. lactalbumin and some proteins secreted in the pancreas (for instance ribonuclease) are glycosylated.

➤ **Oligosaccharide linkages in glycoproteins:**

(a) **O-linked** oligosaccharides contain a glycosidic linkage to the hydroxyl group of Serine or Threonine residues (pink), exemplified here with GalNAc as the sugar at the reducing end of the oligosaccharide. One simple chain and one complex chain are shown.

(b) **N-linked** oligosaccharides contain an *N*-glycosyl bond to the amide nitrogen of an Asn residue (green), exemplified here with GlcNAc as the terminal sugar.



Oligosaccharide linkages in glycoproteins.

➤ **Mucins are glycoprotein components of mucus**

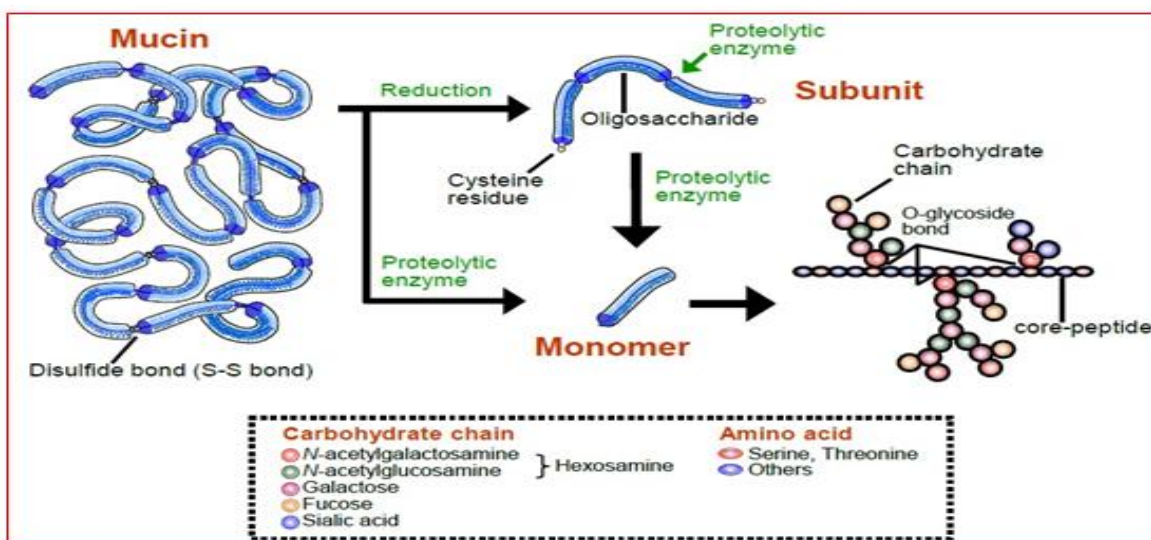
○ **Structure:**

- Mucins (mucoproteins) are secreted or membrane glycoproteins that can contain large numbers of *O*-linked oligosaccharide chains.
- In mucins, the protein component is extensively glycosylated to Serine or Threonine residues by *N*-acetylgalactosamine. Mucins are capable of forming large polymeric structures and are common in mucous secretions.

○ **Significance:**

- Mucins are present in most secretions they give mucus its characteristic slipperiness.

- These glycoproteins are synthesized by specialized cells in the tracheobronchial, gastrointestinal, and genitourinary tracts.
- Because a key function of mucins is to act as a lubricant, mucins are abundant in saliva.
- The important feature of the mucins is a region of the protein backbone termed the variable number of tandem repeats (VNTR) region that is rich in serine and threonine residues that are O-glycosylated. Indeed, the carbohydrate moiety can account for as much as 80% of the molecule by weight.
- Mucins adhere to epithelial cells and act as a protective barrier; they also moist the underlying cells.
- Mucins protect cells from environmental abrasions, such as stomach acid, inhaled chemicals in the lungs and bacterial infections.
- Mucins have roles in fertilization, the immune response, and cell adhesion.
- Mucins are over-expressed in bronchitis and cystic fibrosis, and the over-expression of mucins is characteristic of adenocarcinomas- cancers of glandular cells of epithelial origin.



Polymeric structure of mucin molecules.

3. Glycolipids :-

➤ Structure:

- Glycolipids are membrane sphingolipids in which the hydrophilic head groups are oligosaccharides.

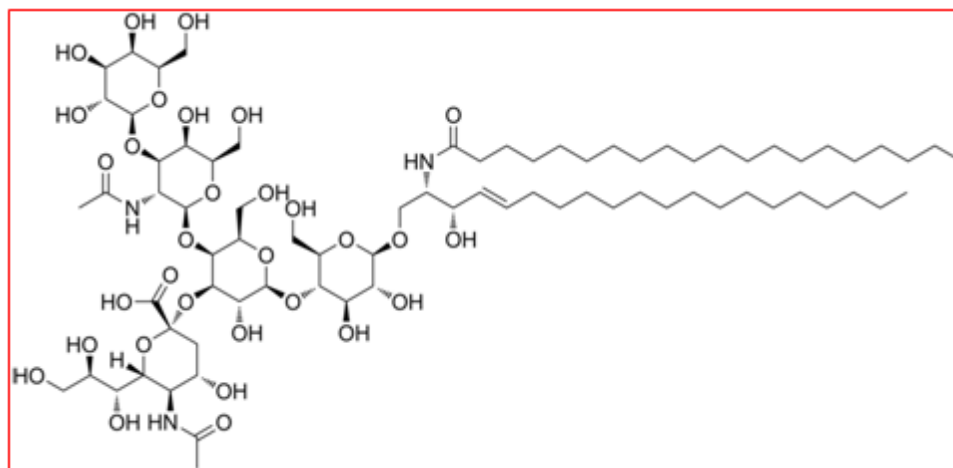
➤ **Significance:**

- As in glycoproteins the oligosaccharides act as specific sites for recognition through protein called lectins.
- The brain and neurons are rich in glycolipids, which help in nerve conduction and myelin formation.
- Glycolipids also play a role in signal transduction in cells.

➤ **Gangliosides: A type of glycolipids**

Structure: These are membrane lipids of eukaryotic cells in which the polar head group, the part of the lipid that forms the outer surface of the membrane, is a complex oligosaccharide containing a sialic acid and other monosaccharide residues.

Significance: Like glycoprotein, a few oligosaccharide moieties of gangliosides determine human blood groups. Like the oligosaccharide residues of glycoproteins, those of membrane lipids are generally found on the outer face of the plasma membrane.

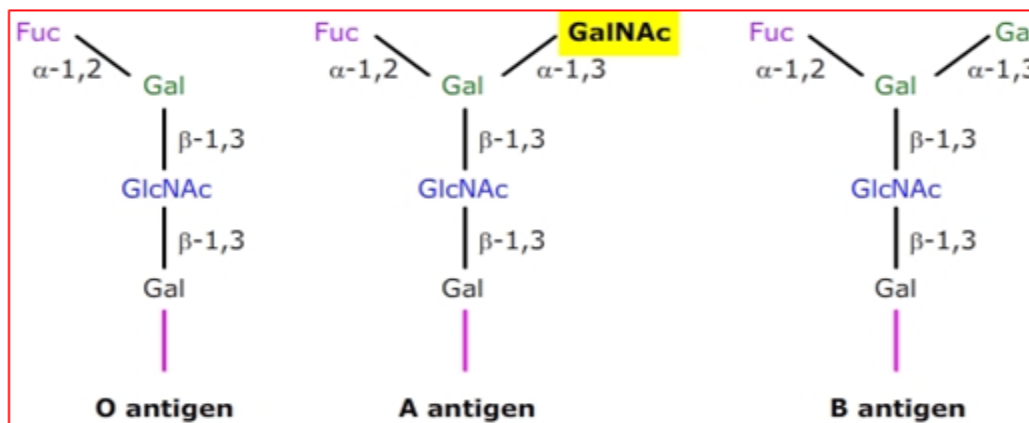


Gangliosides, a glycolipid contain carbohydrate moiety

➤ **Carbohydrate moieties determine the blood group :-**

Carbohydrates are important constituent of blood group system in human blood transfusion. The associated anti-A antibodies and anti-B antibodies are usually immunoglobulin M, abbreviated as IgM antibodies. IgM antibodies are generally synthesized in the first years of life through sensitization to ambient substances for instance food and microbes. The corresponding blood group carbohydrate structures, designated ABH, are found at the termini of oligosaccharide chains on glycoproteins and

glycolipids on the surface of RBC and endothelial & most epithelial cells. The immunodominant monosaccharide that determines blood group A, is a terminal α -1,3-linked N-acetylgalactosamine (GalNAc), whereas the corresponding monosaccharide of blood group B specificity is an α -1,3-linked galactose. Group O cells lack chains, which instead are terminated with α -1,2-linked-fucose.



Structures of A, B and O Oligosaccharide Antigens.

Abbreviations: Fuc, fucose; Gal, galactose; GalNAc, N-acetylgalactosamine; GlcNAc, N-acetylglucosamine.

✚ Lectins: Informational carbohydrate binding proteins

➤ Structure:

- Lectins (from the Latin *legere*, "to select") are specific proteins that bind with informational carbohydrate molecule and mediate many biological processes.
- The diverse carbohydrate structures displayed on cell surfaces are well suited to serve as interaction sites between cells and their environments.

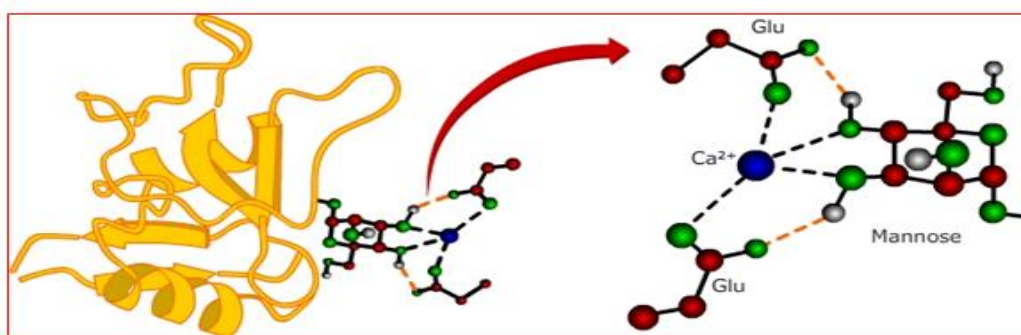
➤ Functions:

- Lectins are ubiquitous, being found in animals, plants, and microorganisms. Certain lectins, for example calnexin, work as chaperones in protein folding.
- Lectins have role in a diverse variety of cell-cell recognition, signaling and adhesion processes and intracellular targeting of newly synthesized proteins.
- Plant lectins are most common in seeds, and act as deterrents to insects and other predators.
- In the laboratory, purified plant lectins are used as reagents for detecting and separating glycans and glycoproteins having different oligosaccharide moieties.

➤ **Classification of lectins:**

Based on their amino acid sequences and biochemical properties Lectins can be categorized into various families.

1. **Intracellular lectins:** These include calnexin family, M-type, L-type and P-type which are found in luminal compartments of the secretory pathway and function in the trafficking, sorting and targeting of maturing glycoproteins.
2. **Extracellular lectins:** These include C-type, R-type and galectins which are either secreted into the extracellular matrix or body fluids, or limited to the plasma membrane and mediate a range of functions such as cell adhesion, cell signalling and pathogen recognition.
3. **C type** (for calcium-requiring) is a large class of lectins found in animals. These proteins commonly contain a domain of 120 amino acids which is responsible for carbohydrate binding.



Structure of a C-type carbohydrate-binding domain of an animal lectin.

A calcium ion joins a mannose carbohydrate moiety to the lectin.

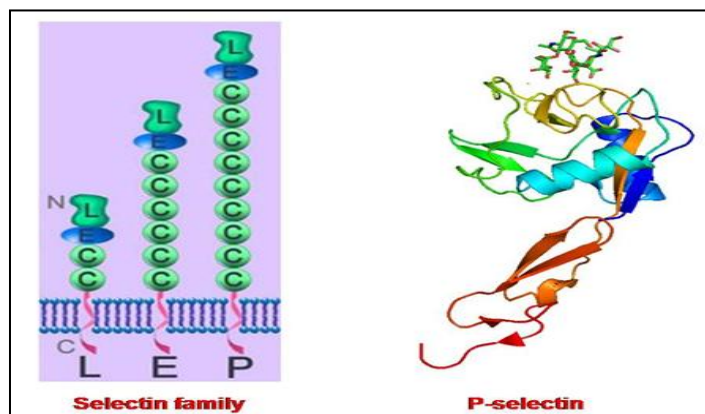
➤ **Selectins: C-type lectins**

Single transmembrane polypeptide selectins occur only in vertebrates, where they undergo calcium dependent binding via their amino-terminal domain to carbohydrates. Selectins are C-type lectins with calcium-dependent carbohydrate recognition domains (CRDs).

Selectin family: The selectin family of lectin mediates function/migration of circulatory cells: **L**-leukocytes; **E**-endothelial cells; **P**-platelets and endothelial cells. Thus selectins have L, E, and P forms bind specifically to carbohydrates on lymph-node vessels, endothelium, or activated blood platelets, respectively.

Structure: The selectin family possess large, highly glycosylated, extracellular domains, a single length transmembrane domain and a small cytoplasmic tail. They contain a single C-type lectin domain (L) at their extracellular amino termini (N).

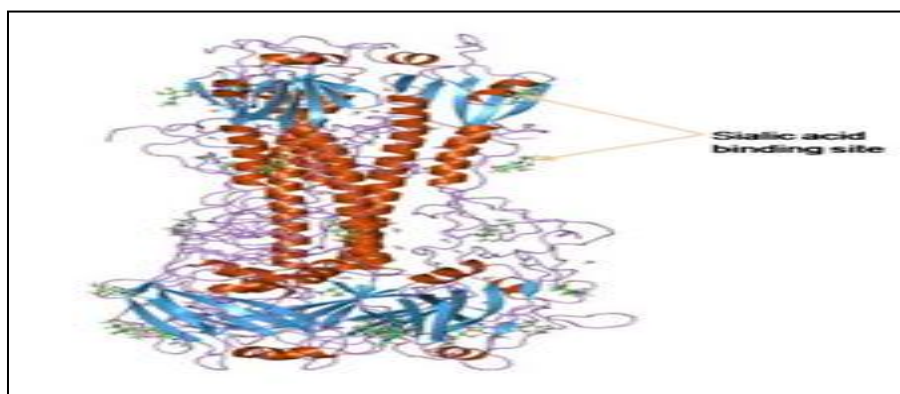
Functions: Selectins bind immune-system cells to the sites of injury in the inflammatory reaction.



Left image: L, E and P form of selectin contains C-type lectin domain (L) at their extracellular amino termini (N), a single spanning transmembrane domain, and a small cytoplasmic tail. Right image: Crystallographic structure of P selectin lectin bound to sugar, shown in sticks.

Hemagglutinin: Viral Protein that binds with carbohydrate of host cell

Influenza virus recognizes **sialic acid moieties** present on cell-surface glycoproteins. The viral protein which binds to these carbohydrates is termed as **hemagglutinin**. After occurrence of these surface interactions and the virus has been entered into the cell, another viral protein, *neuramidase*, cleaves the glycosidic bonds to the sialic acid residues, freeing the virus to infect the cell. Inhibitors of this enzyme could be as one of the potential anti-influenza agents.



Structure of a Part of Influenza Hemagglutinin.

This viral protein has multiple binding sites for linking to sialic acid residues on the target-cell surface.

Glycomics and Glycobiology

Glycomics is the systematic characterization of all of the carbohydrate components of a given cell or tissue, including those attached to proteins and to lipids. For glycoproteins, this also means determining which proteins are glycosylated and where in the amino acid sequence each oligosaccharide is attached. It offers insights into normal patterns of glycosylation and the ways in which they are altered during development or in genetic diseases or cancer. Current methods of characterizing the whole carbohydrate complement of cells depend heavily on sophisticated application of mass spectroscopy.

Glycobiology, is the study of the structure and function of glycoconjugates is one of the most active and exciting areas of biochemistry and cell biology. This field has made us to understand that how cells use specific oligosaccharides to encode important information about intracellular targeting of proteins, cell-cell interactions, cell differentiation and tissue development and extracellular signals.

The composition, properties, and occurrence of the polysaccharide

Summary Structure and role of some polysaccharides and glycoconjugates

Polymer	Type*	Repeating unit	Size (number of monosaccharide units)	Roles
Starch	Homo-	($\alpha 1 \rightarrow 4$)Glc, linear	A few thousand to 500,000	Energy storage: in plants
Amylose				
Amylopectin	Homo-	($\alpha 1 \rightarrow 4$)Glc, with ($\alpha 1 \rightarrow 6$)Glc branches every 24–30 residues	Up to 10^6	Energy storage: in plants
Glycogen	Homo-	($\alpha 1 \rightarrow 4$)Glc, with ($\alpha 1 \rightarrow 6$)Glc branches every 8–12 residues	Varies (several million)	Energy storage: in bacteria, animal cells
Cellulose	Homo-	($\beta 1 \rightarrow 4$)Glc	Up to 15,000	Structural: gives rigidity, strength to plant cell walls
Chitin	Homo-	($\beta 1 \rightarrow 4$)GlcNAc	Very large	Structural: gives rigidity, strength to exoskeletons of insects, spiders, crustaceans
Peptidoglycan	Hetero-, with peptides attached	MurNAc($\beta 1 \rightarrow 4$)GlcNAc	Very large	Structural: gives rigidity, strength to bacterial cell envelope
Glycosaminoglycan (hyaluronate)	Hetero-, acidic	GlcUA($\beta 1 \rightarrow 3$)GlcNAc	Varies ($>10^6$)	Structural: extracellular matrix in skin, connective tissue; viscosity, lubrication in joints of vertebrates
Proteoglycans	Hetero-, with protein attached; largely carbohydrate	Uronic acid ($\beta 1 \rightarrow 3$)-linked with sulfated hexosamine	Varies	Structural: resilience, viscosity, lubrication in joints of vertebrates

* Each polymer is classified as a homopolysaccharide (homo-) or heteropolysaccharide (hetero-).